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Real-time attitude control of large rectangular pipe-jacking machines using a mechanics-informed digital twin

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ABSTRACT

Traditional experience-based methods for controlling the attitude of pipe-jacking machines frequently result in notable deviations from the planned trajectory, posing risks such as excessive jacking forces and structural damage to segments. Achieving precise control of machine attitudes is vital for large-section pipe-jacking projects. This paper presents a mechanics-informed digital twin framework designed to reduce trajectory deviations by leveraging intelligent control systems. A dynamic control platform processes diverse real-time data from construction operations, issues immediate risk alerts and suggest corrective forces derived from a theoretical load model. As demonstrated in a super-large rectangular pipe-jacking project in China, this approach effectively predicted attitude-related moments and provided dependable guidance for trajectory corrections, enhancing construction safety and efficiency. Building on this progress, combining mechanics-informed digital twin approach with machine learning techniques for attitude prediction and control is expected to further enhance the intelligence and adaptability of rectangular pipe-jacking operations.

1. Introduction

Rectangular underground structures offer over 20 % higher spatial utilization compared to circular ones, helping to reduce overall construction costs and providing significant green and low-carbon benefits [1,2]. As a result, rectangular pipe-jacking technology has rapidly developed and is increasingly applied in various urban underground projects [3,4]. To meet growing engineering demands, the size of rectangular pipe-jacking machines has continued to increase, placing higher requirements on control accuracy and construction reliability [5,6]. During pipe-jacking operations, the machine is expected to advance precisely along the Designed Tunnel Axis (DTA). However, due to uneven mass distribution, complex geological conditions, and varying control parameters, the machine often deviates from the DTA, failing to meet required construction tolerances [7–9]. Although operators may frequently perform attitude adjustments, current experience-based control methods rely heavily on manual operations and are prone to delayed responses, making it difficult to effectively reduce tunnel deviation [9–11].

Tunnel deviation leads to increased jacking resistance, which may exceed the machine's designed capacity and cause significant construction risks [12]. Excessive jacking force on the pipe segments can also result in cracking, structural damage, and joint leakage, posing serious safety threats [13]. These challenges are particularly prominent in large-section rectangular pipe-jacking, where multi-cutter designs and heterogeneous soils significantly complicate attitude control [14–16]. Furthermore, during construction near existing structures, deviations in the machine's attitude and position may lead to collision risks. For instance, in twin or multi-tunnel projects, excessive tunnel deviation can widen the gaps between adjacent tunnels, making it difficult to form an integrated structure [17]. Therefore, it is imperative to introduce informative and digital technology to achieve precise and real-time attitude control of rectangular pipe-jacking machines, thereby ensuring construction accuracy, reducing trajectory deviation, and

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enhancing structural safety.

The attitude control of pipe-jacking and shield machines has attracted significant research attention in recent years. To enhance the precision control of the machine's attitude, the force analyses of machine-soil interactions in shield tunnelling have been studied, and the accuracy of proposed theoretical models has been validated through engineering cases [18,19]. While theoretical models have enhanced our understanding of machine–soil interactions, they lack real-time applicability. Additionally, there are substantial differences in the forces acting on rectangular pipe-jacking machines compared to circular shield machines. For example, the multi-cutter design leads to more uneven forces between the front and rear, making the machine prone to nose-diving during construction. The uneven distribution of soil chamber pressure further complicates attitude adjustments for rectangular pipe-jacking machines [5,20]. On the other hand, to improve the real-time attitude control for shield machines, machine learning algorithms have been used to predict the attitude and motion trajectory during shield tunnelling [21]. Shen et al. [22] proposed a framework for shield attitude prediction using a random forest deep learning method, investigating the correlation between input parameter combinations and prediction accuracy. Zhou et al. [11] developed a deep learning model to achieve real-time prediction of shield machine attitude and position under complex environments. Although these studies have improved the real-time attitude prediction for rectangular pipe-jacking construction, aiding operators in making corrective decisions, such decisions remain largely experiential and retain significant uncertainty. Therefore, establishing a correction decision platform based on theoretical force analysis models of the machine, combined with real-time position, attitude, and geological conditions, can provide operators with real-time decision support for corrections and is expected to significantly improve the accuracy of rectangular pipe-jacking machine control [23].

Digital Twin (DT) technology, through the real-time mapping of physical entities and virtual models [24], contributes to intelligent construction and cost-effective management in civil engineering. This technology has been promoted and applied in civil engineering [25–27], providing a new approach for establishing a real-time correction decision platform for rectangular pipe-jacking machines. Wu et al. [28] proposed a digital twin framework based on Bayesian optimization categorical boosting and a multi-objective evolutionary algorithm based on decomposition, which effectively helps to control the attitude of large-diameter slurry shields. Ye et al. [29] developed a digital twin-based platform for smart early warning and risk management, which has been successfully applied in engineering practice. However, the application of digital twins in the field of underground engineering is still in its infancy [30,31], and it has not yet been applied to real-time prediction and auxiliary correction decision-making for the attitude of rectangular pipe-jacking machines.

The application of DT technology in pipe-jacking construction faces challenges similar to those in shield tunnelling: insufficient real-time data acquisition and processing, issues with the immediacy and stability of data transmission, complex interactions among geotechnical, mechanical, and structural components, and difficulties in achieving efficient collaborative decision-making between human operators and intelligent platforms [32]. Zhang et al. [33] proposed a digital twin-based intelligent monitoring and control method for shield tunnelling that utilizes deep learning algorithms to optimize tunnelling parameters; however, their approach does not effectively resolve the issue of real-time attitude correction for shield machines. Moreover, data-driven algorithms like deep learning demand high-quality, high-volume data and often fail to capture the underlying physical mechanisms, making their outputs difficult to interpret or validate and thus less suitable for real-time guidance in shield tunnelling [34,35]. Additionally, the results from these models are typically based on data from specific conditions, which limits their generalisability to other engineering projects [34]. Consequently, researchers have increasingly turned to physics-based models to guide tunnel boring machine operations, a strategy that may better address the challenge of real-time attitude correction for pipe-jacking machines under complex geological conditions [36–38].

This paper explores applying mechanics-informed DT technology to enable the intelligent attitude control of large rectangular pipe-jacking machine. The proposed DT framework significantly enhances the efficiency and accuracy of attitude control by integrating real-time monitoring, risk warning, and recommended correction thrust for the articulated cylinders. The key innovations of this paper are as follows: (1) a mechanics-informed DT framework is proposed to achieve real-time intelligent control of the pipe-jacking machine's attitude; (2) a soil-machine interaction load model is established and integrated into the DT framework, providing a theoretical foundation for attitude control during pipe-jacking construction; (3) A DT-based intelligent control platform is developed, enabling dynamic visualization of the pipe-jacking process, risk warnings, and correction force recommendations through computational algorithms.

The rest of this paper is structured as follows: Section 2 outlines the engineering background. Section 3 details the architecture of the DT-based intelligent control platform for the pipe-jacking machine's attitude. Section 4 explains the theoretical analysis model for the attitude of the rectangular pipe-jacking machine used in this paper. Section 5 presents a case study of the platform's application in a practical tunnel construction project. Section 6 concludes the paper.

2. Engineering background

The Shasan Station, located in the Bao'an District of Shenzhen, China, is a two-story underground island station. As shown in Fig. 1, the two ends of the station were built using the open-cut method, while the middle part of the station, which is 70 m long and passes under a large concrete box culvert, was constructed using the twin closely-spaced rectangular pipe-jacking method [17]. As shown in Fig. 2 (a), the cross-section of the station during construction is 13.53 m high and 22.6 m wide, divided into two small sections: the left-line and right-line substructures, with a close spacing of 5 cm. Each substructure is built in sequence with a super-large rectangular pipe-jacking machine. The left- and right-line substructures are then connected to form an integrated underground space for operational use through structural transformation, as shown in Fig. 2 (b). As shown in Fig. 3, the pipe-jacking machine used is 13.55 m high and 11.295 m wide, comprising two smaller pipe-jacking machines: the upper and lower submachines. The machine is equipped with 45 articulation cylinders to assist with attitude correction, arranged in the middle of the machine. These cylinders are divided into six groups to improve control efficiency, with their parameters listed in Table 1.

During rectangular pipe-jacking, the machine typically exhibits a snake-like motion (see Fig. 4(a)). Therefore, the primary challenge in constructing subway stations using the closely-spaced pipe-jacking method lies in the high-precision control of the machine's attitude and alignment. This is especially critical when the right-line tunnel is jacked closely to the left-line tunnel. Additionally, for super-large rectangular pipe-jacking machines, the uneven distribution of soil strata within the excavation area can easily cause an unbalanced force on the machine, exacerbating its deviation. Deviation of the machine towards the left-line tunnel can easily lead to collision risks, while deviation away from the left-line tunnel can widen the gap between the two tunnels, making it difficult to form an integrated station structure (see Fig. 4(b)). Vertical misalignment of the machine can also cause elevation differences between the left and right tunnels, leading to water leakage and reducing the station's usable space, thereby severely impacting the station's functionality. According to design requirements, the net distance between the left- and right-line tunnels must be controlled within 10 cm. However, it is difficult to effectively control the machine's attitude within the design requirements if using the traditional feedback-based control method [2,11]. Therefore, a digital twin approach for the attitude control of rectangular pipe-jacking machines is considered,

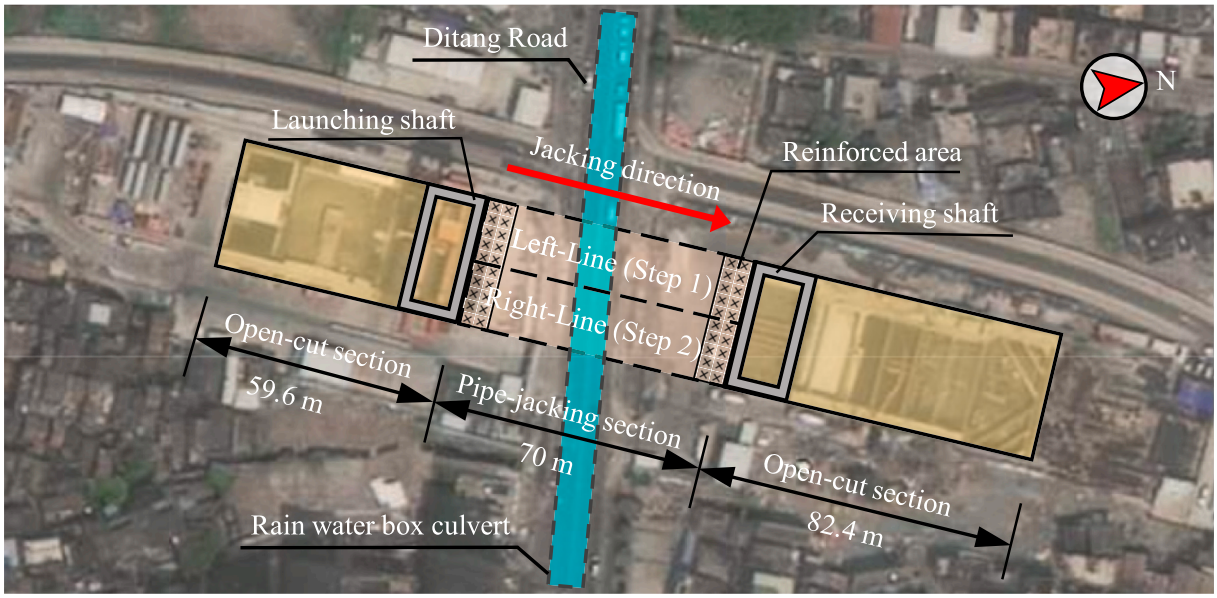


Fig. 1. Plan layout of the Shasan Station.

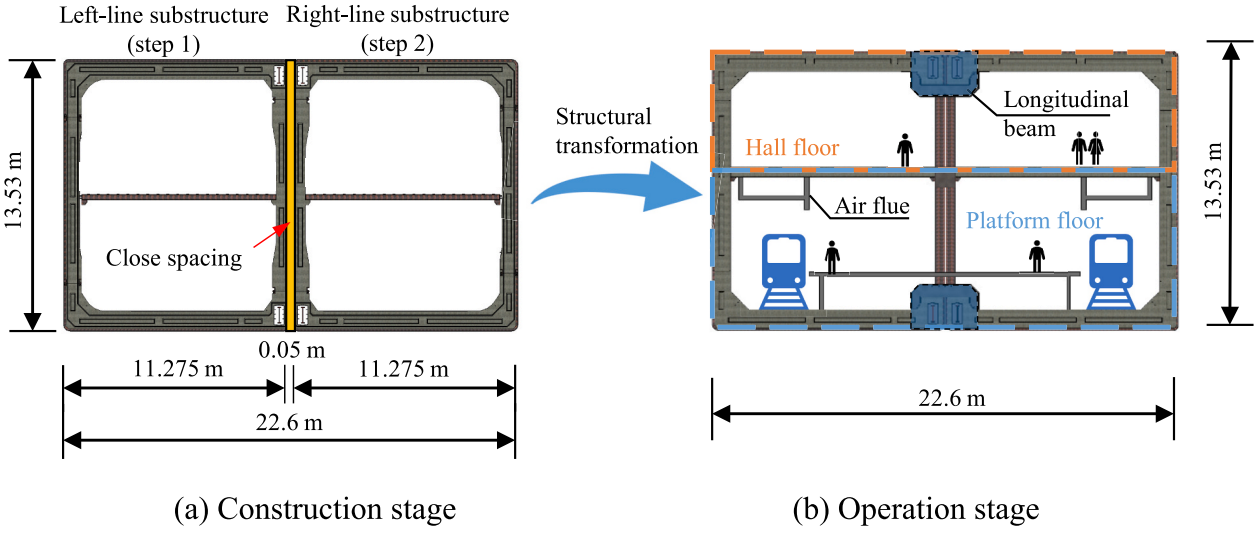


Fig. 2. Profile of the Shasan Station's pipe-jacking section.

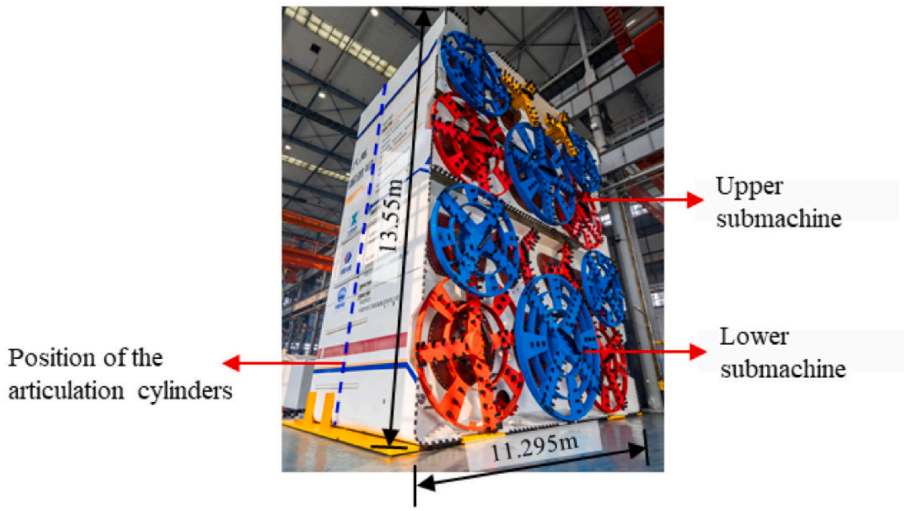
which could help realize real-time risk warnings and intelligent control of the machine, improving the accuracy of attitude control and meeting construction safety requirements.

3. Digital-twin framework for the intelligent control of machine's attitude

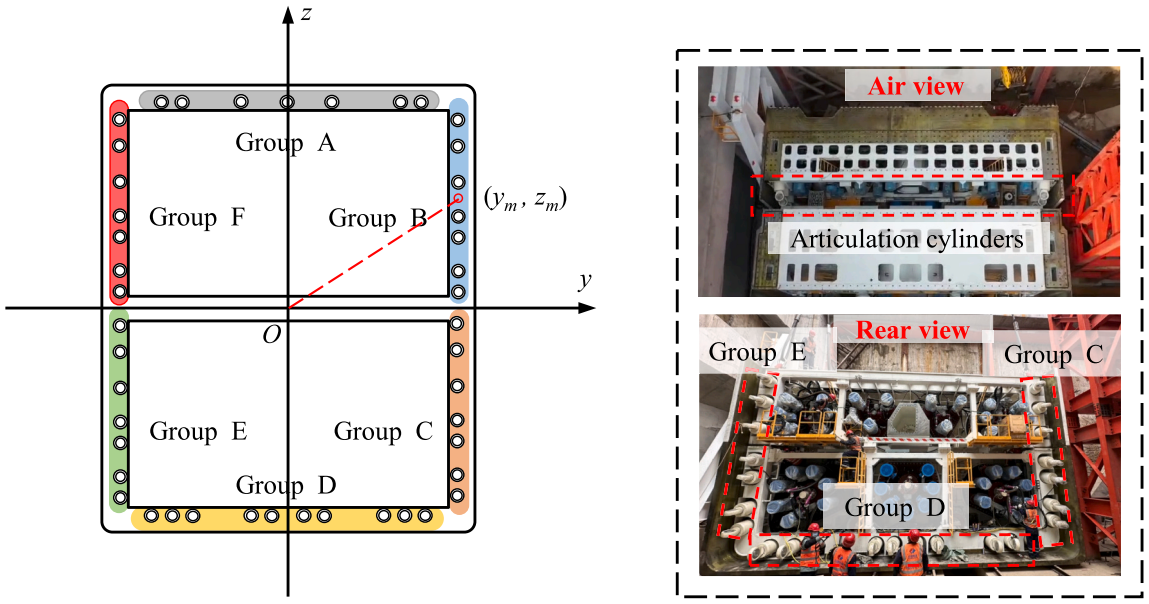
To achieve intelligent attitude control in pipe-jacking operations, this section presents a digital twin (DT) framework that integrates real-time data acquisition, virtual modeling, and intelligent decision-making. The framework consists of a physical layer, a digital layer, and a service layer, enabling real-time interaction between physical systems and virtual models. Section 3 details the functions of each layer and their roles in supporting accurate attitude prediction and control.

3.1. Framework design

To reduce the trajectory deviation of the rectangular pipe-jacking machine, this paper developed an intelligent control platform based on DT technology. According to the DT concept, which comprises five components: physical entity, virtual entity, connections, data, and services [8,39]. The platform's key functions include data visualization, risk warning, and intelligent decision-making for deviation correction. The framework design of this platform is illustrated in Fig. 5. By integrating geological information, various monitoring data of the machine, and other relevant details, the platform aids managers in real-time site information management and provides risk warnings and guidance for timely attitude adjustments by users. The platform operates as follows: first, monitoring data from the pipe-jacking construction is automatically collected through a multi-function collection base station and various sensors. This information is then wirelessly transmitted from the base station to the data center and the platform. A virtual model of the tunnel construction, including the pipe-jacking machine, surrounding soil, and tunnel structure, is created, and dynamic mapping between the construction site and the virtual model is achieved based on the collected data. With the input of dynamic data, the attitude of the rectangular pipe-jacking machine can be simulated using the established theoretical analysis model (introduced in Section 4). As a result, the



(a) Super-large rectangular pipe-jacking machine [17]



(b) Distribution of the articulation cylinders (c) Articulation cylinders of the lower submachine

Fig. 3. Super-large rectangular pipe-jacking machine and the articulation cylinders.

Table 1
Designed parameters of the articulated cylinders.

Parameter	Value
Number of the articulated cylinders	45
Groups	6
Stoke of the articulated cylinders	300 mm
Maximum rectifying force	160,314 kN
Maximum rectifying angle	Up and down: 1.2° Left and right: 1.5°

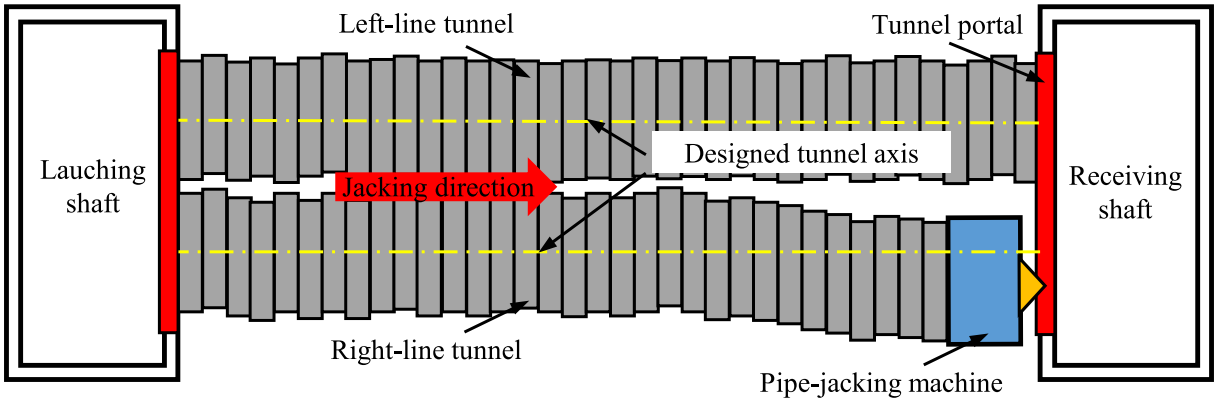
platform enables parallel deduction and intelligent decision-making, providing automatic attitude prediction, risk warnings and guidance parameters (such as the correction angle, thrust adjustment values). Real-time risk warnings are visually displayed for managers, and guidance for machine deviation rectification based on risk assessment is provided to operators for precise adjustments. Ultimately, the platform offers a dynamic, interactive, and intelligent approach to controlling the attitude of the rectangular pipe-jacking machine.

3.2. Physical layer

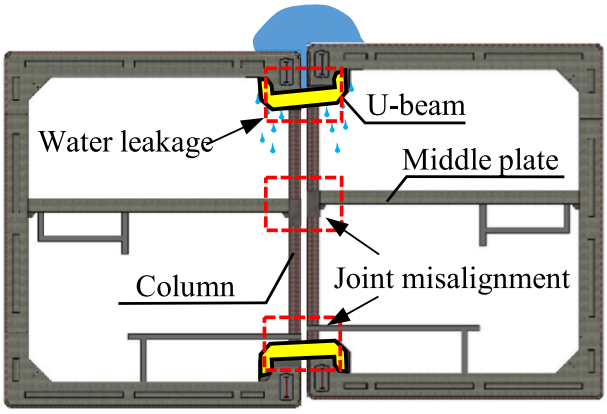
The physical layer serves as the foundation of the DT system, focusing on the real-time collection and transmission of key construction data. This section outlines its structure and core functions.

3.2.1. Multi-information of pipe-jacking construction

Pipe-jacking construction is a complex process involving various types of information, including geological data, tunnel structure details, and specifications of the pipe-jacking machine. To effectively reduce trajectory deviation, it is essential for the platform to collect and integrate these diverse data types in a timely manner, thereby providing a fundamental information source for accurate control and adjustment.



(a) Diagram of misalignment of twin closely-spaced tunnel



(b) Risks induced by tunnels misalignment

Fig. 4. Risks induced by misalignment of twin closely-spaced tunnels.

Since tunnel structural data are less utilized in this paper—primarily obtained from design drawings for constructing the BIM model—the discussion mainly focuses on geological data and pipe-jacking machine data.

Regarding data acquisition, geological data were primarily obtained through borehole sampling, while pipe-jacking machine data were collected via the tunnelling system and guidance system and transmitted in real-time through wireless devices.

In terms of data storage, geological data were stored in a constructed database supporting export formats such as Excel documents. The coordinates of borehole points and corresponding stratum information were recorded, with coordinate indexing enabling real-time retrieval based on the jacking distance. Similarly, pipe-jacking machine data were stored in a corresponding database, recording real-time data from various systems of the machine, with timestamp indexing to facilitate real-time access.

For data processing, to improve the retrieval efficiency of geological data, the original data were preprocessed by segmenting the geological model into smaller zones along the pipe-jacking direction. A weighted algorithm was applied to generate stratum information for each zone, enabling efficient real-time access. As for pipe-jacking machine data, preprocessing methods such as data extraction and outlier detection were employed to ensure data validity.

3.2.2. *Real-time data acquisition and transmission*

To effectively acquire construction site information, real-time data collection is achieved through multiple base stations with distinct functions, all connected to a central gateway that manages and controls terminal equipment. Various sensors, including those integrated into different systems of the pipe-jacking machine and the automatic guidance system, are linked to the collection base stations to gather real-time data from the construction site. Data are collected at the second-level interval, ensuring real-time acquisition of operational parameters. The collection base station is responsible for aggregating diverse information during pipe-jacking, while the transmission base station handles both uplink and downlink data transmission and control.

The hardware layout at the construction site is illustrated in Fig. 6. Sensors and theodolites are strategically installed on the construction site, and laser targets are positioned on the pipe-jacking machine. Most of the sensors installed on the machine are connected to the control platform via wire transmission, which ensures high transmission speed and stability [32]. Additionally, these sensors are calibrated both in the factory and on-site following assembly, further enhancing the reliability and stability of the data and effectively resolving issues related to real-time data acquisition and transmission. Collected data is transmitted to the collection base station via a wireless link and subsequently sent to the data center through the transmission base station. Theodolites, laser targets of the automatic guidance system, and sensors on various systems of the pipe-jacking machine are installed and calibrated based on the specific conditions of the tunnel construction. Additionally, manual checks of the machine's position and attitude are performed at regular intervals to verify the data measured by the automatic guidance system. To ensure uninterrupted operation of the acquisition system, a continuous power supply is maintained, and relevant equipment is routinely

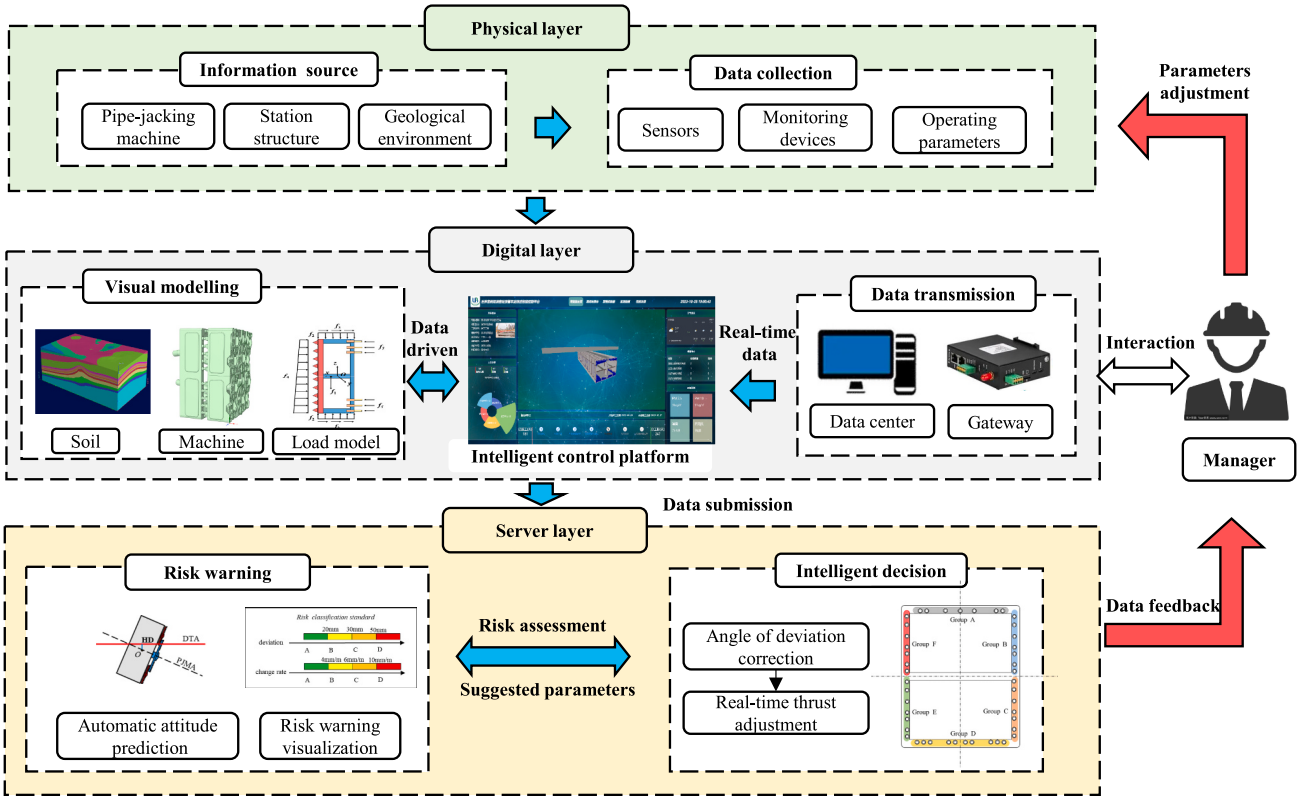


Fig. 5. DT framework for intelligent attitude control of machine's attitude.

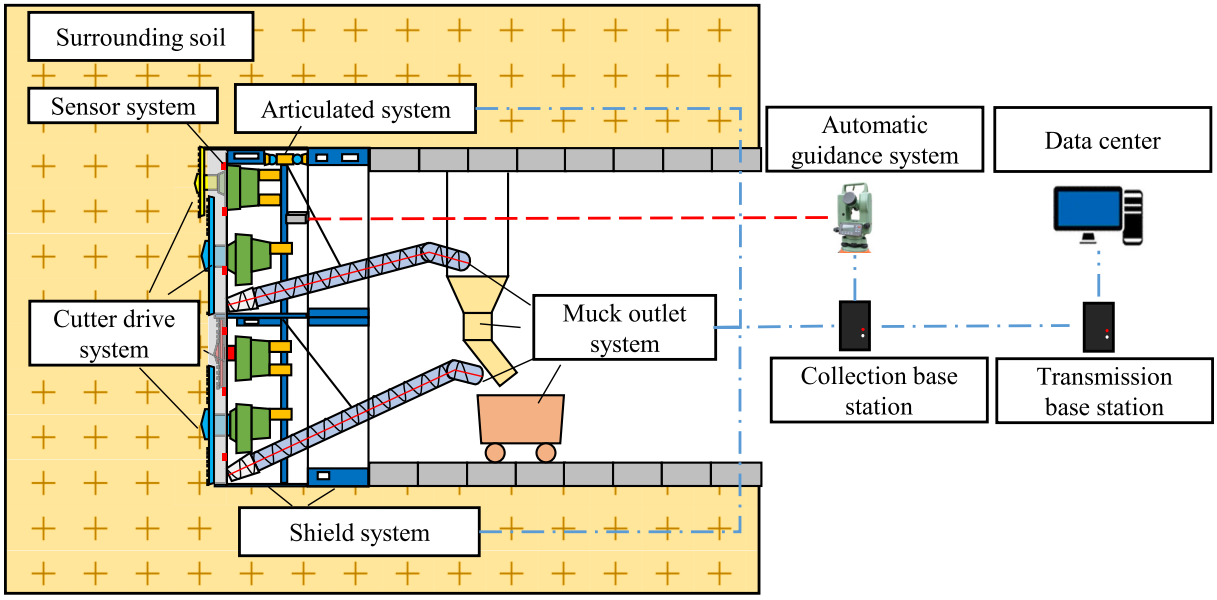


Fig. 6. Real-time data transmission and layout of hardware devices.

inspected and serviced. This setup allows real-time transmission of monitoring data to the platform, enabling operators to access critical information through the Internet.

3.3. Digital layer

To support real-time monitoring and intelligent control of the machine's attitude during pipe-jacking, the digital layer constructs a virtual simulation environment that integrates data and models for enhanced decision-making.

3.3.1. Digital twins of pipe-jacking construction

The virtual static model of tunnel construction, including the pipe-jacking machine, surrounding soil, and tunnel structure, is developed using Geographic Information System (GIS) technology and Building Information Modeling (BIM) technology before pipe-jacking construction. For example, ArcGIS software was utilized to construct a 3D geological model based on borehole data, which was integrated into the

front-end of the DT platform for real-time visualization of the geological environment. The geological information around the pipe-jacking machine (e.g., soil distribution, mechanical parameters) was extracted from the established 3D geological model and used as input data for the platform's real-time calculations, supporting dynamic optimization of the machine's attitude correction. During the construction process, dynamic mapping between the construction site and the virtual model is achieved through real-time monitoring data, which facilitates the updating of the virtual model's information.

In this framework, the intelligent control platform is divided into two core components: the front end and the back end, based on a distributed Linux architecture to enhance usability for operators and managers. The back end utilizes the lightweight Flask framework for Web services and implements load balancing through Nginx to manage high concurrency and heavy traffic requests. For databases, InfluxDB and MySQL are appropriately configured, with index optimization to improve query efficiency. The temporal database InfluxDB is employed to manage real-time monitoring data, while the relational database MySQL handles non-temporal data. On the front end, the Vue.js progressive framework is selected, combined with Element-Plus for page layout design, and Three.js for the dynamic loading of 3D models, such as geological models. Thus, models of the pipe-jacking machine, tunnel structure, surrounding soil, and environment are established. Information from various stages of pipe-jacking construction, including investigation, design, and construction, is input and recorded. In the digital layer of this platform, the attitude of the rectangular pipe-jacking machine can be simulated by integrating the theoretical analysis model (see Section 4) with real-time data.

3.3.2. Real-time interaction

The real-time data and 3D models of pipe-jacking construction displayed on the platform, as shown in Fig. 7, are synchronized with the construction process through the data acquisition system. Multiple monitoring points are established, including those for the attitude of the pipe-jacking machine and the earth pressure at the excavation face. Users can access the platform via the network to obtain real-time monitoring data. Furthermore, when the monitoring data exceeds predefined warning thresholds, the platform will display information regarding the risk warning level and the corresponding response plans. Based on this risk warning mechanism and response plan, the platform's computer system can automatically generate guidance parameters for the intelligent control of the machine's attitude, which are then provided to operators in the control room.

3.4. Service layer

The service layer utilizes theoretical load model and real-time data to support functions such as predictive simulation and intelligent control. This section introduces how these services assist operators in making data-informed decisions for accurate machine attitude adjustment.

3.4.1. Parallel deduction service

To aid users in assessing the risk warning levels, a risk warning system has been integrated into the platform's software, designed according to current specifications [7]. As shown in Fig. 8, the risk warning workflow utilizes two control indexes: the pipe-jacking machine's deviation and its change rate. Notably, if the sign of the change rate is opposite to that of the deviation, it indicates that the deviation of the pipe-jacking machine will decrease, and the risk level associated with the change rate is categorized as green. The pipe-jacking machine's deviation is determined based on real-time monitoring data, while its change rate is influenced by the machine's attitude.

The risk warning level is determined using these two control indexes. Users can preset standard or threshold values for the deviation and change rate risk levels in the software according to relevant risk management specifications. Consequently, the platform can provide real-time risk warnings based on the monitoring data, which are displayed in red, orange, yellow, or green to indicate different risk levels, as shown in Fig. 9.

As shown in Fig. 10, the pipe-jacking construction process is divided into two stages: the stoppage and advancement stages. During the advancement stage, the pipe-jacking machine moves forward, and soil extracted during excavation is conveyed to the muck bucket. The machine will stop when the muck bucket is full, and the muck bucket will then be transported to the ground surface to unload the muck. Fig. 10 (c) illustrates that parallel deduction is performed during the stoppage stage. Based on the geological conditions and construction parameters from the previous advancement stage, the attitude of the pipe-jacking machine for the next stage is predicted using the theoretical analysis model (introduced in Section 4). This allows the operator to evaluate risks and make timely adjustments. The predicted risk warning level for the next advancement stage is determined in conjunction with the machine's deviation, and intelligent decisions are made based on the predicted risk level.

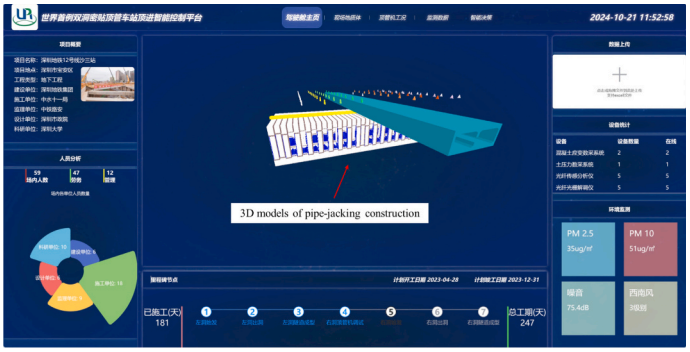
3.4.2. Intelligent decision service

As illustrated in Fig. 10 (c), intelligent decision-making is based on the predicted risk warning level and the corresponding response plan, which guides the intelligent control of the pipe-jacking machine's attitude for the following advancement stage. The workflow of the response plan is detailed in Table 2, where the parameters are selected based on the Chinese pipe-jacking specification [40]. Depending on the risk warning level, various response strategies are employed, including adjustments to monitoring frequency and deviation correction angles. The attitude angle of the machine for the next advancement stage is determined to minimize trajectory deviation based on the correction angle and attitude angle from the previous stage. The relationship between various machine attitude angles and rectification moments can be determined using a load model that accounts for soil-machine interaction. This model leverages sensor data, including soil chamber pressure, articulation cylinder thrust, and geological distribution parameters, to compute the correction force in real time, as discussed in Section 4. Consequently, guide parameters for intelligent attitude control, such as the rectification moments for the articulation cylinders arranged in groups, are calculated automatically using the theoretical analysis model based on the attitude angle. These parameters are then displayed on the intelligent decision interface of the platform. Operators in the control room can use these guide parameters to precisely adjust the articulation cylinders, thereby achieving accurate control of the machine's attitude.

4. Theoretical load model of rectangular pipe-jacking machine

In order to ensure accurate management of the rectangular pipe-jacking machine's attitude, a theoretical load model of the front shield has been proposed [18,41]. The forces impacting the front shield can be classified into three kinds: (a) the weight of the front shield, denoted as $\boldsymbol{f}_1$, which remains constant during pipe jacking; (b) the interactional forces between the front shield and the adjacent soil, including $\boldsymbol{f}_2$, $\boldsymbol{f}_3$ and $\boldsymbol{f}_4$; and (c) the thrust exerted by the articulation cylinders, referred to as $\boldsymbol{f}_5$, which can be actively controlled by the operator. As shown in Fig. 11, $\boldsymbol{f}_1$ represents the weight of the front shield; $\boldsymbol{f}_2$ indicates the normal force exerted on the front shield periphery by the soil; $\boldsymbol{f}_3$ represents the tangential force; $\boldsymbol{f}_4$ indicates the loads affecting the excavation face; $\boldsymbol{f}_5$ signifies the thrust from the articulation cylinders.

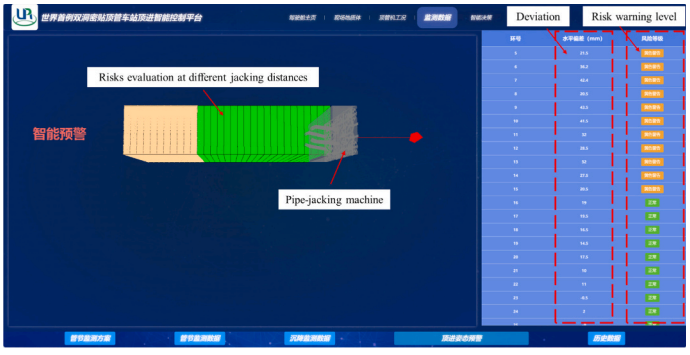
Considering the considerable stiffness of the front shield in comparison to the surrounding soil, it can be effectively treated as a rigid entity. To pinpoint the direction of the forces acting on it, a local coordinate system $\{O\text{-}xyz\}$ has been established, as depicted in Fig. 12. This system $\{O\text{-}xyz\}$ moves with the front shield, with the origin O marking the shield's midpoint. The x-axis is aligned with the machine's designed tunnel axis, the y-axis stretches out horizontally, and the z-axis



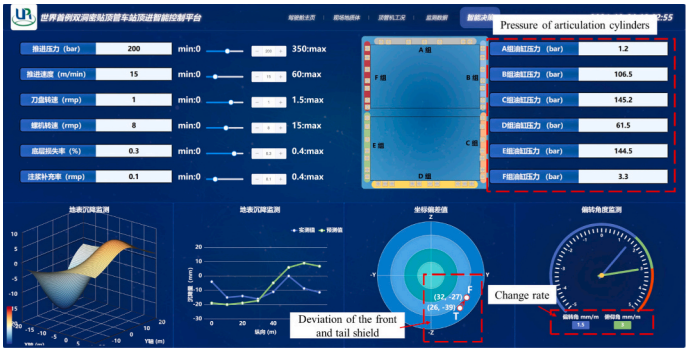
(a) 3D models of pipe-jacking construction



(b) Real-time monitoring data



(c) Risk warning information



(d) Parallel deduction and intelligent control

Fig. 7. Interface of the intelligent control platform.

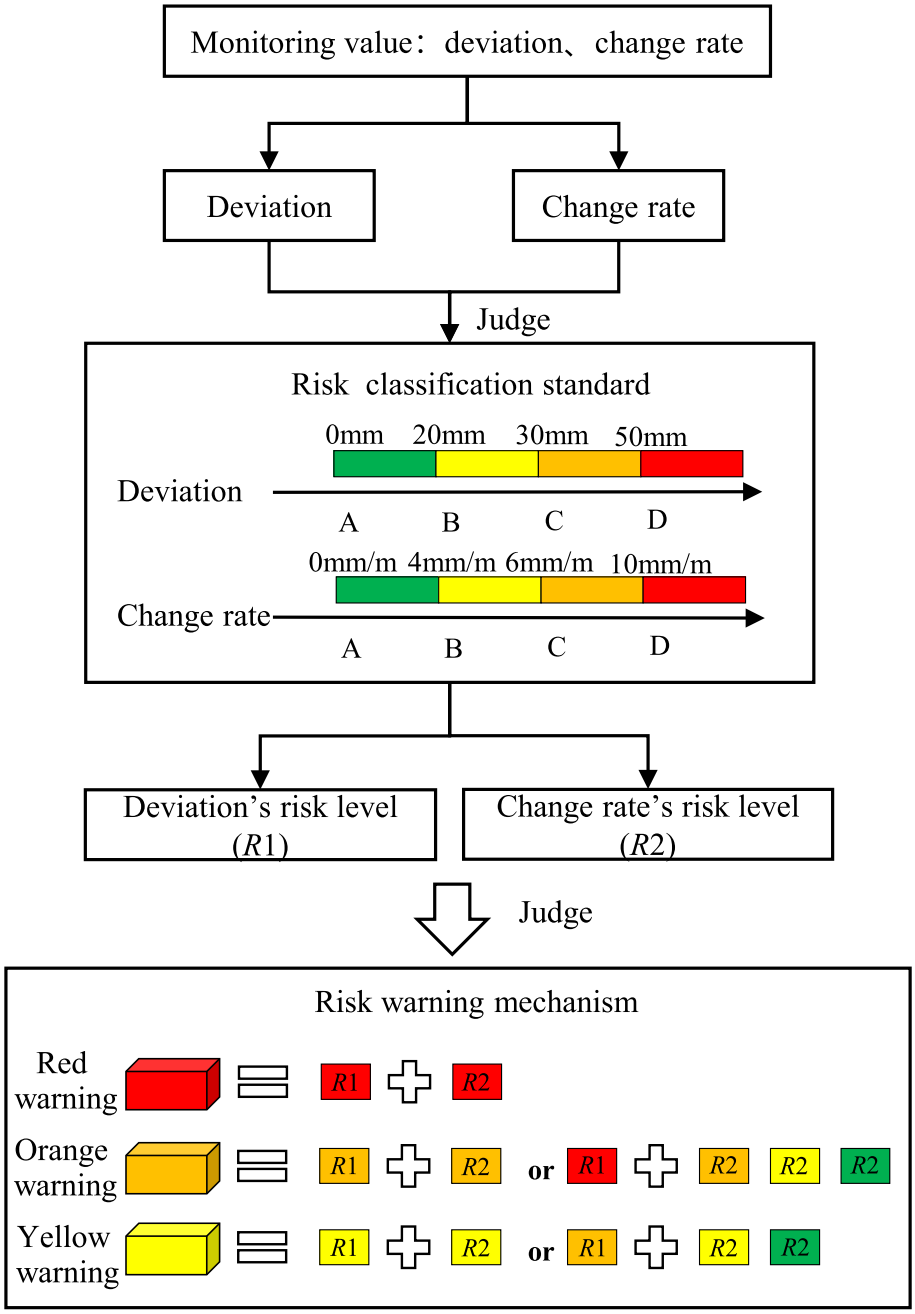


Fig. 8. Workflow of the three-level warning system.

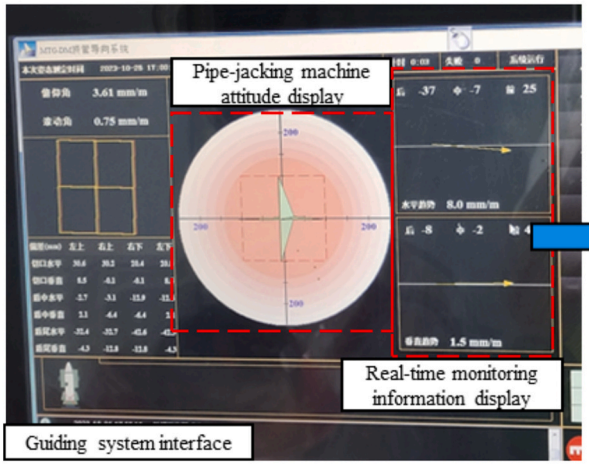
extends vertically. Consequently, the shield's position and attitude during pipe jacking are described by six degrees of freedom. Among these, three pertain to attitude angles: yaw, pitch, and roll. The remaining three are position coordinates, which include advancement distance, Horizontal Deviation (*HD*), and Vertical Deviation (*VD*), as shown in Fig. 13. A positive yaw angle signifies a right turn, while a negative one indicates a left turn. The horizontal deviation is right for positive, left for negative. The pitch angle is positive with upward tilt, negative with downward tilt. The vertical deviation includes positive upward and negative downward. Additionally, the roll angle in a rectangular pipe-jacking machine is generally small and will not be the primary focus of this paper.

4.1. Soil-machine interaction

Within the three load types, the forces interacting between the machine and the adjacent soil ($f_2 \sim f_4$) are intricately linked to various factors, including soil characteristics, operational variables, the machine's attitude and its geometric parameters, posing it difficulties in precisely controlling the machine's position and attitude during pipe jacking. The forces exerted on the front shield periphery mainly comprise the normal force (f_2), the tangential force (f_3), and the moments associated with these forces. It's these forces that predominantly influence the machine's trajectory. Nonlinear springs are evenly distributed along the front shield to model its interaction with the soil. [10,41].

4.1.1. Shield periphery coordinate transformation

During pipe jacking, frequent adjustments of the machine's attitude is necessary to steadily correct trajectory deviations. These adjustments can induce displacement between the machine and the adjacent soil, causing variations in earth pressure across different regions, which has a



Current time:	2023-10-26 17:00:16
Sampling frequency:	12 s per time
Horizontal direction	
Current deviation :	-7.0 mm
Change rate :	(+)8.0 mm/m
Risk warning level:	Without warning
Vertical direction	
Current deviation:	-2.0 mm
Change rate:	(+)1.5 mm/m
Risk warning level:	Without warning

Fig. 9. Example of real-time construction data synchronization and automatic risk warning.

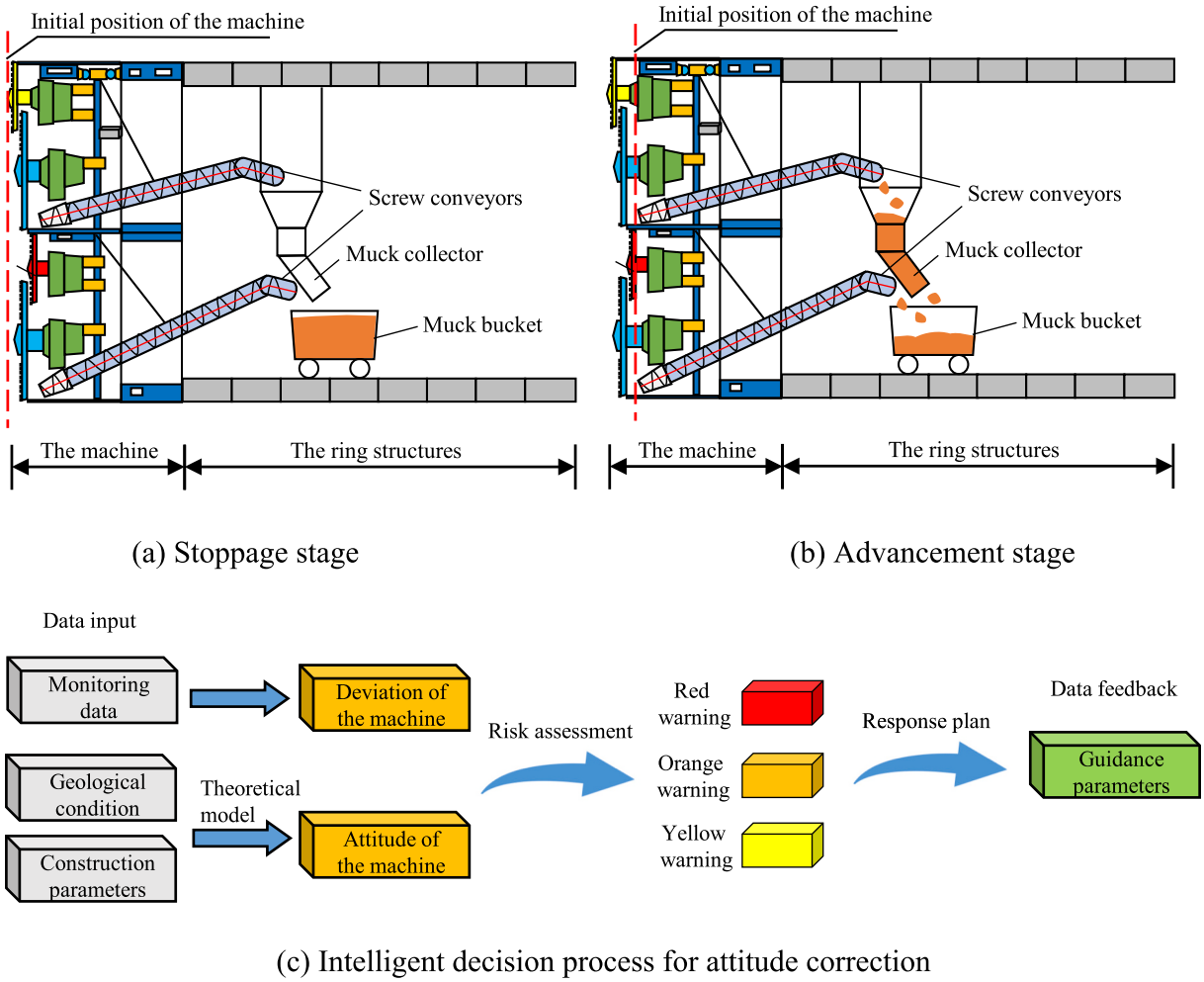


Fig. 10. Intelligent decision process for attitude correction based on construction stages.

considerable effect on the loads f_2 to f_3. Therefore, it is crucial to establish the machine's position and attitude variation prior to calculating the loads f_2 to f_3.

The attitude angles describe the orientation of the front shield. Accordingly, the rotation matrix can be expressed as follows.

$$R_z(\alpha) = \begin{bmatrix} \cos\alpha & \sin\alpha & 0 \\ -\sin\alpha & \cos\alpha & 0 \\ 0 & 0 & 1 \end{bmatrix} \quad (1)$$

Table 2
Workflow of response strategies.

Risk warning level	Monitoring frequency	Deviation angle correction	Other measures
Red warning	Manual retest per 0.3 m	0.3–0.5° per time	
Orange warning	Manual retest per 0.5 m	0.15–0. 3° per time	If necessary, stop construction
Yellow warning	Manual retest per 1.0 m	< 0.15° per time	

$$R_y(\beta) = \begin{bmatrix} \cos\beta & 0 & -\sin\beta \\ 0 & 1 & 0 \\ \sin\beta & 0 & \cos\beta \end{bmatrix} \tag{2}$$

$$R_x(\psi) = \begin{bmatrix} 1 & 0 & 0 \\ 0 & \cos\psi & \sin\psi \\ 0 & -\sin\psi & \cos\psi \end{bmatrix} \tag{3}$$

$$R(\alpha,\beta,\psi) = R_z(\alpha)R_y(\beta)R_x(\psi) \tag{4}$$

where α, β, and ψ denote the yaw, pitch, and roll angles, respectively.

Given that the guidance system can provide the positions of preset monitoring points of the front shield in real-time, it becomes possible to obtain the yaw, pitch, and roll angles of the front shield by analyzing the relative positions of the measurement points. Consequently, the front shield's position and attitude are readily determined.

The variation in the front shield's attitude induces corresponding displacements in both the shield's periphery and the adjacent soil. To accurately calculate the relative displacement, it is essential to first establish the position variation of the shield's periphery. In the coordinate system $\{O\}$, the position vector for a point S located on the shield's periphery can be represented as follows.

$$r_{os} = [x \ y \ z]^{\mathrm{T}} \tag{5}$$

where x, y and z represent the coordinates of the point S.

After attitude adjustment, the change in position vector Δr_{os} caused by attitude adjustment can be determined by.

$$\Delta r_{os} = (R(\alpha,\beta,\psi) - 1)\, r_{os} \tag{6}$$

Consider a section of the front shield as illustrated in Fig. 14. The starting position of this section is indicated by a solid rectangle, whereas the intended new position following the attitude adjustment is represented by a dotted rectangle. The horizontal and vertical displacement of point S, resulting from previous attitude adjustment, can be calculated as follows.

$$\Delta u_{Sh} = \Delta r_{os} \cdot j \tag{7}$$

$$\Delta u_{Sv} = \Delta r_{os} \cdot k \tag{8}$$

where $j = [010]^{\mathrm{T}}$ represents the unit vector along the y axis and $k = [001]^{\mathrm{T}}$ represents the unit vector along the z axis. Δu_{Sh} and Δu_{Sv} represent the horizontal and vertical displacement of point S, which differ from the normal displacement of the soil u_n. Therefore, it is essential to determine u_n.

As illustrated in Fig. 15, when $y = \pm b/2$ and $\Delta u_{Sh} y > 0$, or $z = \pm H_0/2$ and $\Delta u_{Sv} z > 0$, the point S falls within the passive state zone, resulting in a positive value for u_n; when $y = \pm b/2$ and $\Delta u_{Sh} y = 0$, or $z = \pm H_0/2$ and $\Delta u_{Sv} z = 0$, the point S falls within the static state zone, leading to a zero value for u_n; when $y = \pm b/2$ and $\Delta u_{Sh} y < 0$, or $z = \pm H_0/2$ and $\Delta u_{Sv} z < 0$, the point S falls within the active zone, which means u_n takes on a negative value. Accordingly, u_n can be solved as follows.

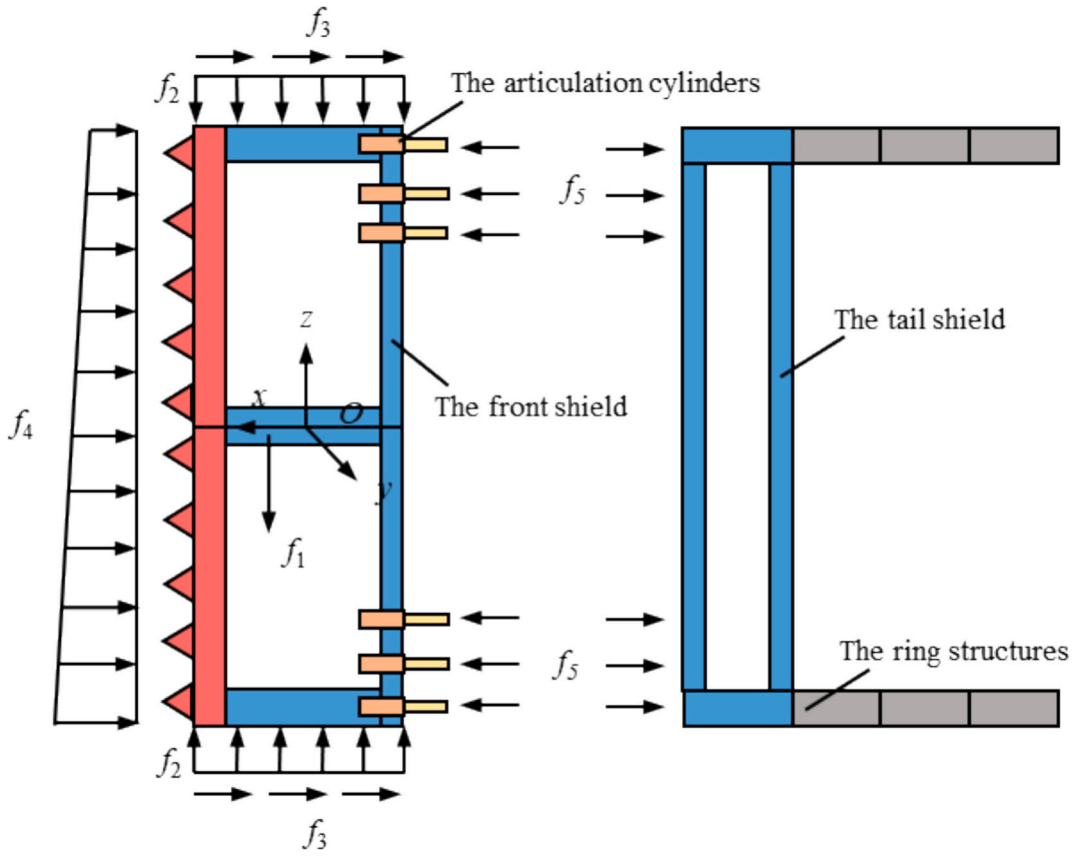


Fig. 11. Load model of rectangular pipe-jacking machine.

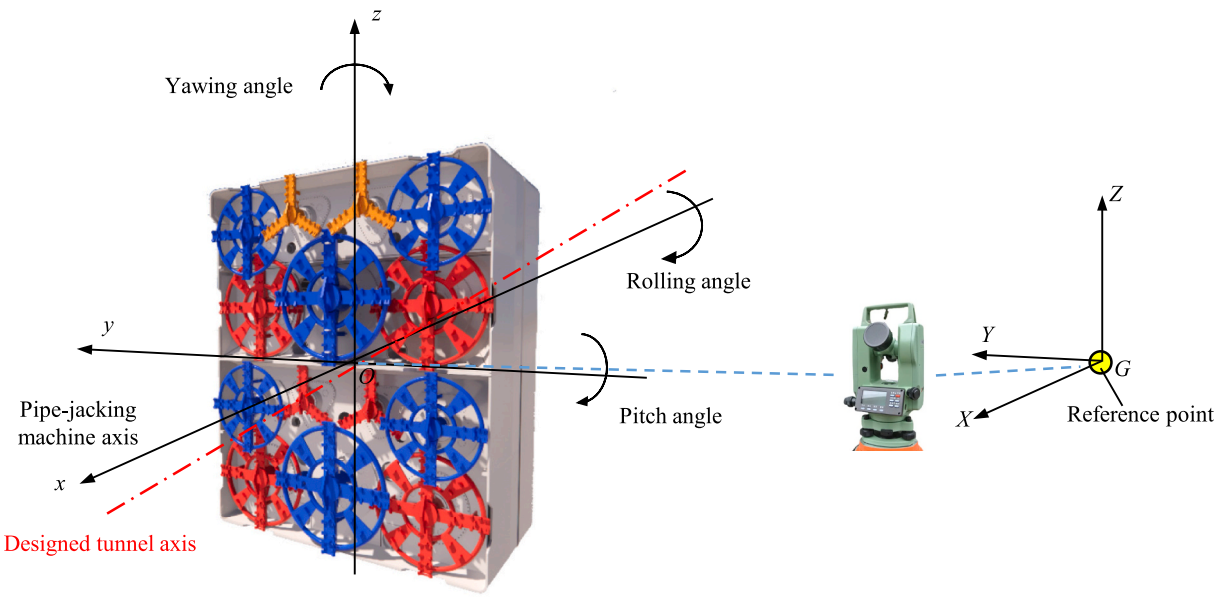


Fig. 12. Attitude angles of rectangular pipe-jacking machine.

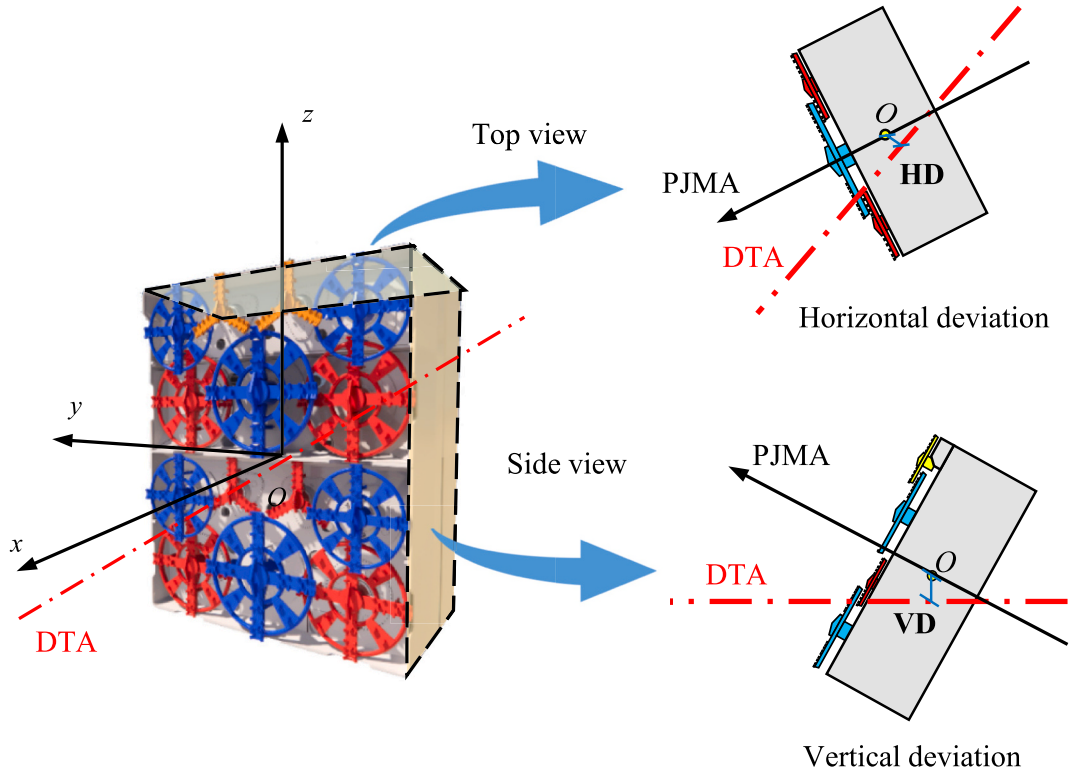


Fig. 13. Deviation of rectangular pipe-jacking machine.

$$u_n = \begin{cases} |\Delta u_{Sh}| \cdot \mathrm{sign}(\Delta u_{Sh} y), y = \pm b/2 \\ |\Delta u_{Sv}| \cdot \mathrm{sign}(\Delta u_{Sv} z), z = \pm H_0/2 \end{cases} \quad (9)$$

where b and H_0 represent the machine's width and height.

It can be observed that u_n at point S is determined by rotation matrix R, the front shield parameter H_0 and b, and position parameters x, y, z of point S. Therefore, u_n is represented as a multivariate nonlinear function.

$$u_n = f(R, H_0, b, x, y, z) \quad (10)$$

4.1.2. Normal interaction between shield and soil

In order to assess the loads $\boldsymbol{f}_2$ to $\boldsymbol{f}_3$, as depicted in Fig. 16, the front shield periphery is divided into many small units. The force acting on each unit is represented by calculating the force at its central point. The overall interaction is then determined by summing the forces of all the units.

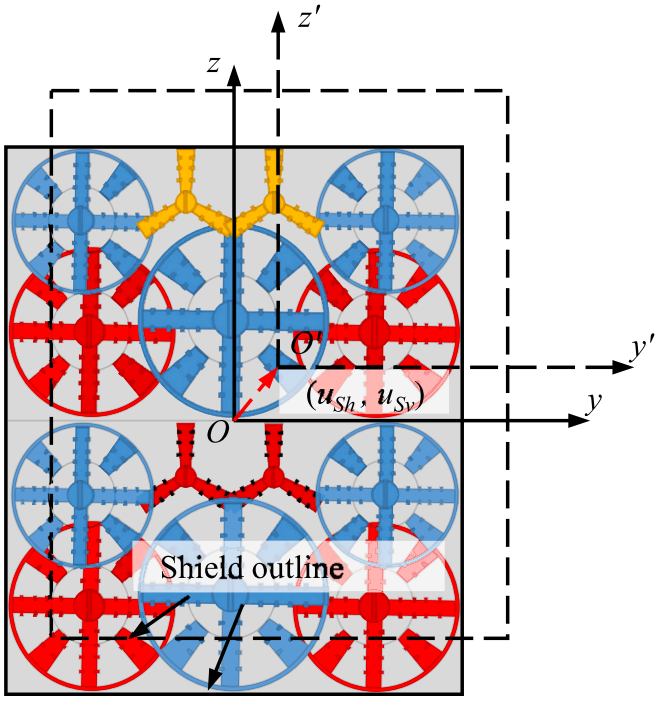


Fig. 14. Schematic diagram of shield displacement.

Before adjusting the machine's attitude, the front shield is in its initial state, with normal earth pressure around its periphery in a balanced condition. This pressure can be determined by referring to the earth's static pressure. Particularly, the pressure exerted on the top plate is calculated using Eq. (11), the pressure on the bottom plate is calculated using Eq. (12), and the horizontal earth pressure on the side plates is calculated using Eq. (13).

$$\sigma_{v0} = K_{v0}\gamma h \tag{11}$$

$$\sigma_{v0} = K_{v0}\gamma h + \frac{G}{A} \tag{12}$$

$$\sigma_{h0} = K_{h0}\gamma h \tag{13}$$

where σ_{v0} represents the vertical earth pressure at rest, while σ_{h0} indicates the horizontal earth pressure at rest; K_{v0} represents the coefficient of vertical earth pressure at rest, and it's set to one; K_{h0} represents the coefficient of horizontal earth pressure at rest; γ represents the soil's unit weight; G represents the front shield's weight; A represents the area of bottom plate of the front shield; h represents the overlying soil's thickness.

However, the normal earth pressure will change after the attitude adjustment. As shown in Fig. 17, the sum of $\boldsymbol{f}_2$ can be determined using ground reaction curves. The curves are represented as follows [10,41].

$$K_i(u_n) = \begin{cases} (K_{io} - K_{imin})\tanh\left[\dfrac{a_i u_n}{K_{io} - K_{imin}}\right] + K_{io} & (u_n \le 0) \\ (K_{io} - K_{imax})\tanh\left[\dfrac{a_i u_n}{K_{io} - K_{imax}}\right] + K_{io} & (u_n \ge 0) \end{cases} \tag{14}$$

where $i = v$ or h represents either vertical or horizontal directions; the subscript o represents the static condition; min and max refer to the active and passive state of the earth pressure, respectively; a_i represents the slope of the earth pressure coefficient curves in the static condition, which is defined as $a_i = k_i / \sigma_{v0}$; where k_i represents the coefficients of subgrade reaction; u_n signifies the soil's normal displacement when the computed point is in the compaction region, $u_n > 0$, and when it is in the extension region, $u_n < 0$.

The normal earth pressure (σ_n) at point S can be solved by.

$$\sigma_n = K_i(u_n)\sigma_{v0} \tag{15}$$

where $K_i(u_n)$ represents the earth pressure coefficient at any point S.

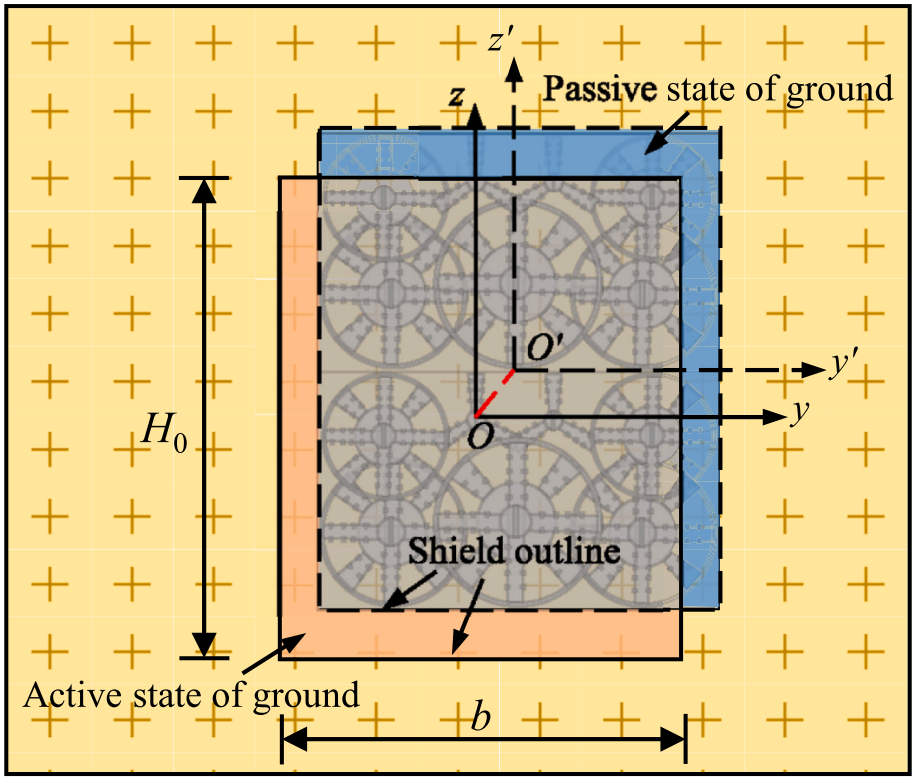


Fig. 15. Normal displacement of adjacent soil.

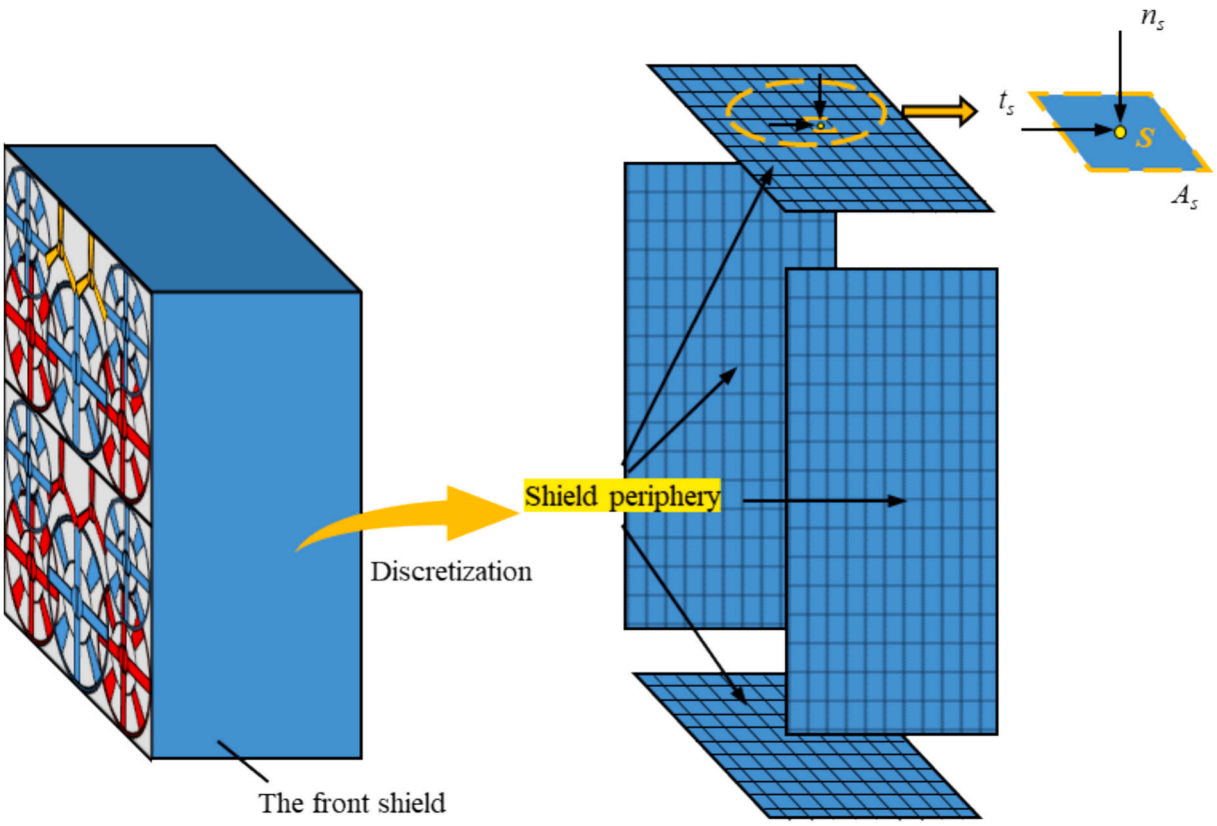


Fig. 16. Discretization of shield periphery.

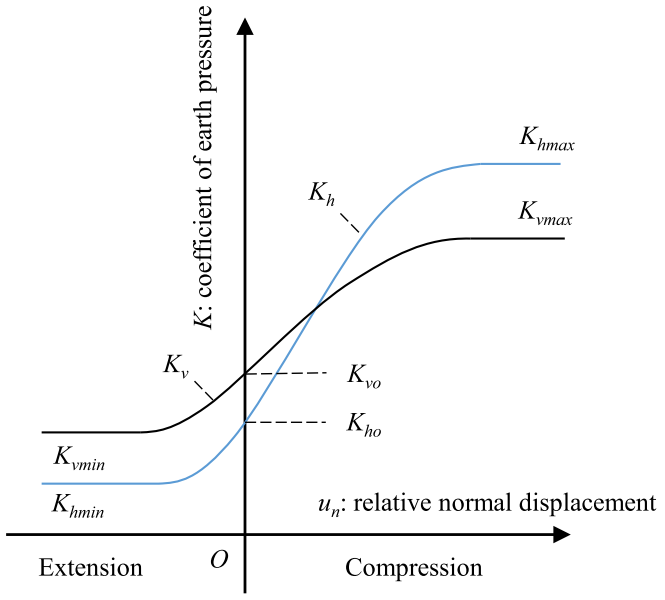


Fig. 17. Ground reaction curve. [10,41].

Then the normal force f_2 can be calculated by.

$$f_2 = \sum \sigma_n A_s n_s \tag{16}$$

where A_s indicates the grid area containing point S; n_s represents the unit normal vector at point S.

The normal earth pressure (σ_n) will result in associative moment at origin point O that can be calculated as follows.

$$M(f_2) = \sum \sigma_n A_s (r_{os} \times n_s) \tag{17}$$

4.1.3. Tangential interaction between shield and soil

The tangential interaction between front shield periphery and soil will also influence the attitude control during pipe jacking. Fig. 18 illustrate the machine-soil friction curve, which can be divided into the ascending section and the platform section. The tangential stress τ_s [27] can be solved as.

$$\tau_s = \mu \sigma_n \tag{18}$$

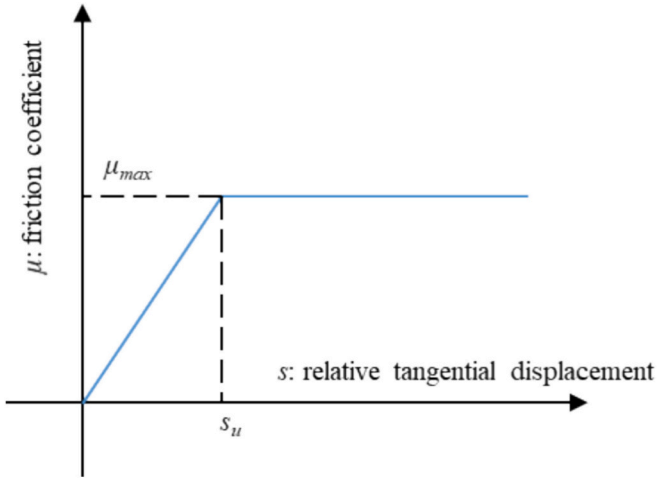


Fig. 18. Friction curve of machine and soil.

$$\mu = \begin{cases} \frac{s}{s_u}\mu_{\max} & 0 \le s \le s_u \\ \mu_{\max} & s \ge s_u \end{cases} \tag{19}$$

where τ_s denotes the tangential stress at point S; σ_n denotes the normal stress at the same point; μ denotes the friction coefficient at the interface between the machine and the soil; $\mu_{\max} = 0.4\tan\varphi$, where φ represents the friction angle of the soil; s denotes the relative tangential displacement at point S; s_u denotes the relative tangential displacement when tangential stress reaches its maximum.

Subsequently, the tangential force $\boldsymbol{f}_3$ exerted on the front shield is represented as.

$$f_3 = \sum \tau_s A_s t_s \tag{20}$$

where t_s denotes the unit tangential vector at point S.

The tangential stress τ_s will result in associative moment at origin point O that can be expressed as.

$$M(f_3) = \sum \tau_s A_s (r_{os} \times t_s) \tag{21}$$

4.1.4. The loads acting on the excavation face

During the pipe-jacking process, the front shield encounters earth pressure in front of cutter. In the case of large rectangular pipe-jacking machines, the Soil Chamber (SC) pressure distribution is more complex, which influences the machine's attitude. As illustrated in Fig. 19, the machine's soil chamber used in Shasan Station is segmented into four distinct areas, designated as SC1-SC4. A total of 19 earth pressure sensors are affixed to the soil chamber's rear panel, tagged from TY6 to TY24, to monitor earth pressure variations in SC. The mean earth pressure for different areas, labeled P1 to P4, is derived from the monitoring data provided by these sensors, as reported in the previous study [6]. The load exerted on the excavation face ($\boldsymbol{f}_4$) is determined by summation of the four uniform surface loads (P1 to P4). The forces and associative moments are calculated as follows.

$$f_4 = \left[-\sum_{i=1}^{4} P_i\, h_i\, b \;\; 0 \;\; 0 \right] \tag{22}$$

$$M(f_4) = \left[0 \;\; \sum_{i=1}^{4} P_i\, h_i\, b\, z_i \;\; 0 \right] \tag{23}$$

where $\boldsymbol{M}(\boldsymbol{f}_4)$ denotes the associative moment led by f_4; P_i denotes the mean earth pressure across SC1 to SC4; h_i denotes the height for SC1 to SC4; z_i denotes the central coordinates along the z-axis for SC1 to SC4.

4.2. Other loads exerted on the machine

The remaining forces exerted on the front shield includes the weight of the front shield ($\boldsymbol{f}_1$) and the thrust of the articulation cylinders ($\boldsymbol{f}_5$). Due to the uneven mass distribution of the machine, the front shield mass is mainly concentrated in the cutters, the f_1 and associative moment ($\boldsymbol{M}(\boldsymbol{f}_1)$) at origin point O can be calculated by.

$$f_1 = [0 \;\; 0 \;\; -G] \tag{24}$$

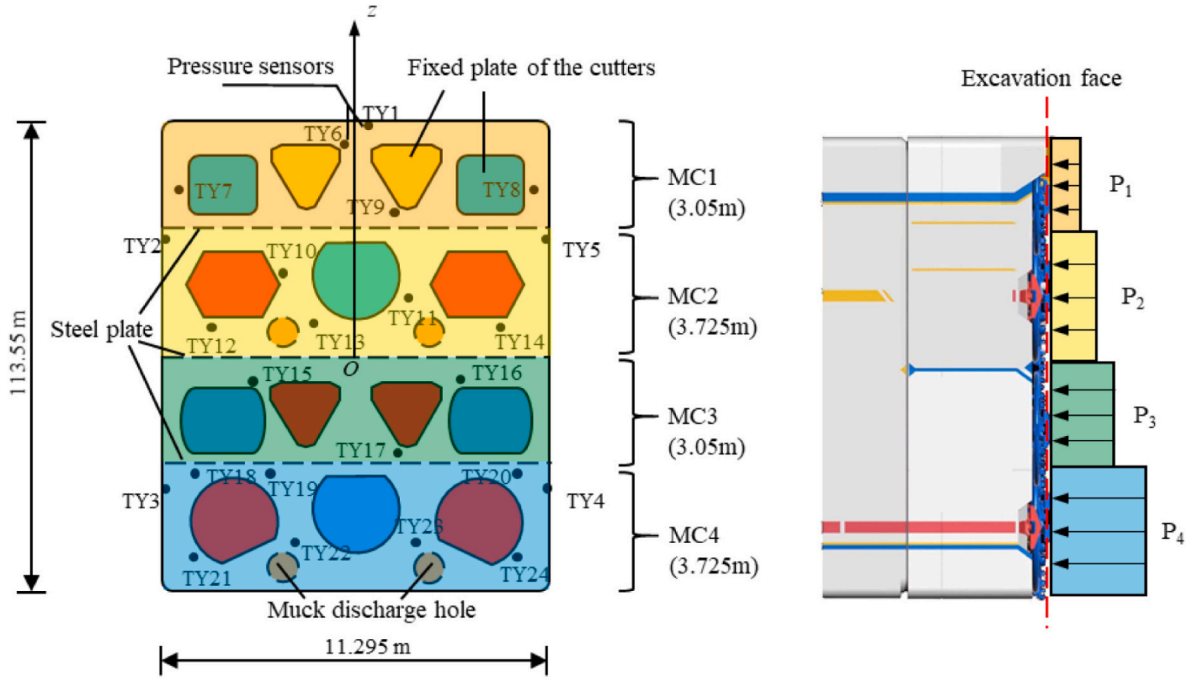
$$M(f_1) = [0 \;\; -G{\cdot}e \;\; 0] \tag{25}$$

where e denotes the distance between the front shield's centroid and its geometric center.

The machine's articulation cylinders are controlled in groups (see Fig. 3), with each cylinder within the same group applying the same force. Accordingly, the total thrust of all articulation cylinders ($\boldsymbol{f}_5$) can be expressed as.

$$f_5 = \left[\sum_{m=1}^{6} t_m\, n_m \;\; 0 \;\; 0 \right] \tag{26}$$

where m represent the count of the articulation cylinders groups; t_m represents the unit thrust of the corresponding articulation cylinders group; n_m represents the total number of cylinders within that group.



(a) Distribution of the pressure sensors (b) Soil chamber pressure distribution

Fig. 19. Distribution of loads exerted on excavation face.

The associative moment ($\boldsymbol{M}(\boldsymbol{f}_5)$) at origin point O induced by $\boldsymbol{f}_5$ can be calculated by.

$$M(f_5) = \left[0 \ -\sum_{m=1}^{6} t_m n_m z_m \ -\sum_{m=1}^{6} t_m n_m y_m\right] \tag{27}$$

where z_m denotes the unit coordinates of the corresponding group on the z axis; y_m denotes the unit coordinates of the corresponding group on the y axis. The central coordinate positions of each articulated cylinder group are listed in Table 3.

4.3. Calculation of the machine's attitude

The pipe-jacking machine advances slowly when jacking in soil, and the front shield essentially maintains a state of stasis The static equilibrium equation for the front shield is expressed by.

$$\sum_{n=1}^{5} f_{ne} = 0 \tag{28}$$

$$\sum_{n=1}^{5} M(f_{ne}) = 0 \tag{29}$$

The expansion of Eqs. 28 and 29 are as follows.

$$f_{1e} + f_{2e} + f_{3e} + f_{4e} + f_{5e} = 0 \tag{30}$$

$$M(f_{1e}) + M(f_{2e}) + M(f_{3e}) + M(f_{4e}) + M(f_{5e}) = 0 \tag{31}$$

where $e = x$, y, or z, represents the various directions of different loads.

Based on the established load model Section 4, all the forces acting on the front shield during pipe jacking can be calculated. This model allows for determining the relationship between attitude angles (α, β, ψ) and the rectification moments ($\boldsymbol{M}(\boldsymbol{f}_{5y})$, $\boldsymbol{M}(\boldsymbol{f}_{5z})$) generated by $\boldsymbol{f}_5$, as shown in Fig. 20. The theoretical analysis model incorporates operating parameters, attitude parameters, soil parameters, geometric parameters, and their interrelationships. This comprehensive model provides a solid foundation for parallel deduction and intelligent decision-making (introduced in Section 3.4) regarding the rectangular pipe-jacking machine's attitude management.

5. Case study and validation

To assess the efficacy of the suggested intelligent control platform that integrates the load model, the left-line tunnel of Shasan Station was chosen as a case study. The main parameters are presented in Table 4. The data acquired by the sensors of the machine are initially stored in the local database of the intelligent control platform. Subsequently, the data processing module embedded in the control platform performs preprocessing operations, including data extraction, outlier detection, and data transformation, to generate a refined dataset. The refined dataset is then fed into the embedded prediction algorithm, which is designed based on the load model of the pipe jacking machine. The predicted attitude parameters are validated against predefined safety thresholds and instantaneously transmitted to the risk warning module and intelligent decision control unit for auxiliary attitude correction.

Table 3
Central coordinates of articulated cylinder groups.

Articulated cylinder group	n_m	y_m(m)	z_m(m)
Group A	7	0.00	6.30
Group B	7	5.15	3.30
Group C	7	5.15	−3.48
Group D	10	0.00	−6.30
Group E	7	−5.15	−3.48
Group F	7	−5.15	3.30

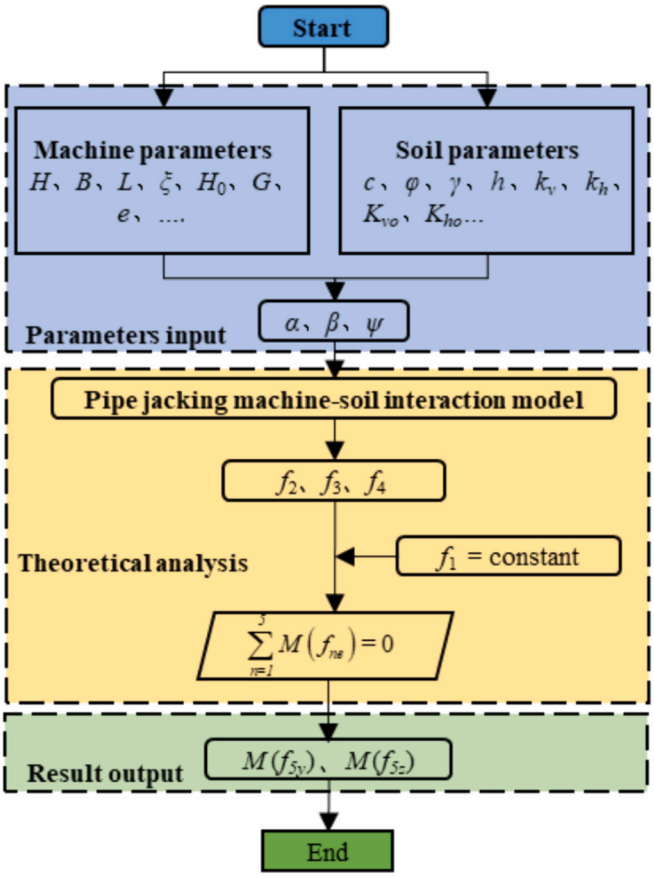


Fig. 20. Calculation procedure for rectification moment.

Table 4
Description of the main operation parameters.

Number	Parameter type	Parameter	Unit
1	Soil parameters	Density	kN/m^3
2		Cohesion	kPa
3		Friction angle	°
4	Operating parameters	Cofficient of subgrade reaction	MPa/m
5		Chamber pressure	bar
6		Cover depth	m
7		Advance speed	mm/min
8		Cutter head rotation speed	r/min
9		Cutter torque	KN·m
10		Total thrust force	kN
11		Pressure of cylinder group A	kPa
12		Pressure of cylinder group B	kPa
13		Pressure of cylinder group C	kPa
14		Pressure of cylinder group D	kPa
15		Pressure of cylinder group E	kPa
16		Pressure of cylinder group F	kPa
17	Shield attitude	Pitch angle	mm/m
18		Roll angle	mm/m
19		Yaw angle	mm/m
20	Shield deviation	Horizontal deviation	mm
21		Vertical deviation	mm

Specifically, by embedding this model into the intelligent platform, the platform can output shield attitude based on input soil and operating parameters, enabling risk warnings. Additionally, it can input shield attitude (such as rectification angles calculated from response strategies) and other parameters to output predicted rectification moments, thereby facilitating intelligent decision-making.

5.1. Data base

The geological profile of the left-line tunnel is shown in Fig. 21. Based on the geological survey findings, the overlying strata at the construction site consist of plain fill and silty clay, while the underlying strata are sandy clay, fully weathered mixed granite, and highly weathered granite. Due to these uneven strata, controlling the attitude of the pipe-jacking machine is challenging, and the machine tends to deviate from the DTA. Detailed mechanical and physical characteristics of the strata have been presented in Table 5, which were primarily obtained from the detailed investigation phase of the Shasan Station project through borehole exploration. Parameters concerning the rectangular pipe-jacking machine and the left-line tunnel are detailed in Section 2.

Operation parameters of the machine were collected using various sensors during pipe jacking, as illustrated in Section 3.2. Attitude parameters of the pipe-jacking machine were collected using an automatic guidance system. These data were transmitted to the data center of the platform. Subsequently, several data preprocessing methods, including data extraction, outlier detection, and data transformation, were applied to the operating and attitude parameters of the machine. The data was processed centrally every minute during the jacking process, and the preprocessing typically took less than 1 s. No significant delays were observed during the data processing on the platform.

5.1.1. Data extraction

The operating and attitude parameters from the advancement stage were collected for preprocessing. Since the machine stopped during muck outward transportation stage, the advancement process was discontinuous. Consequently, these parameters during the stoppage stage were set to zero. First, the data for the advancement stage were extracted from the monitoring data. Advancement Distance (Ad) was used to identify the pipe-jacking construction stage in the operating parameters, which was recorded at a regular intervals Δt. If $Ad(t+\Delta t) - Ad(t) = 0$, the data belong to stoppage stage; otherwise, the data belongs to the advancement stage.

The data from the advancement stage were used as the initial dataset for preprocessing, which included outlier detection and data transformation. However, the first 2.5 % and last 2.5 % of the data, recorded during the transition between the advancement and stoppage stages, were excluded due to their high volatility.

Table 5
Soil parameters of the case project.

Soil name	Density	Cohesion	Friction angle	Cofficient of subgrade reaction	
	(kN/m^3)	(kPa)	(°)	(MPa/m)	(MPa/m)
	γ	c	φ	K_v	K_h
Fill	17.8	15.0	13.0	10	12
Slity clay	19.0	25.0	12.5	18	25
Mud soil	15.9	10	4	5	7.5
Sandy clay	18.5	25.0	22.5	35	45
Fully weathered mixed granite	18.8	30.0	25.0	60	75
Highly weathered mixed granite	19.2	35	27	100	120

5.1.2. Outlier detection

Outliers are some extreme values that markedly differ from the trends observed in the data after preprocess. Due to sensor malfunctions and unstable construction conditions, outliers can represent errors in the monitoring data, often characterized by sudden and substantial increases or decreases. Consequently, it is essential to identify outliers using statistical methods. In this paper, the Interquartile Range (IQR) was employed to detect these anomalies.

5.1.3. Data transformation

During the machine advancement stage, operational and attitude parameters were recorded at 1-s and 12-s intervals. To standardize these parameters for the same construction process, average values corresponding to specific advancement distances were calculated. In this paper a short construction process was defined as the machine advancing a distance of 10 mm. The average values for this process were

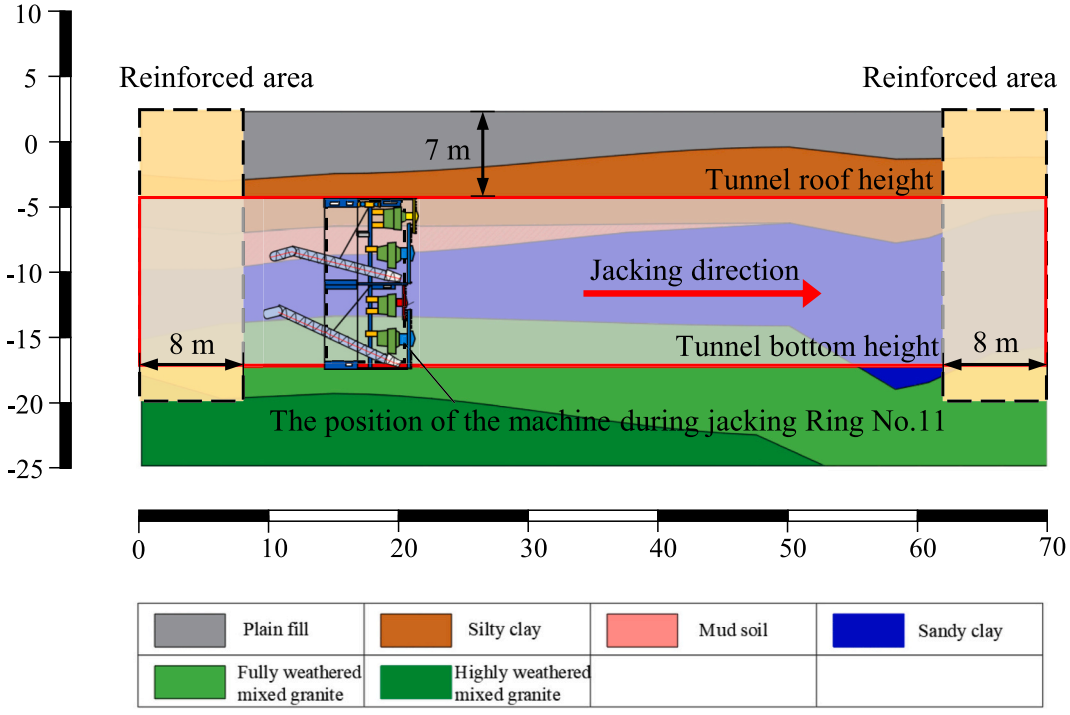


Fig. 21. Geological profiles of the case tunnel.

determined by taking the weighted average of the recorded parameters. Thus, the operating and attitude parameters were averaged according to their respective recording intervals.

$$Op = \frac{\sum_{i=1}^{n} Op_i \cdot \Delta t_{Oi}}{\sum_{i=1}^{n} \Delta t_{Oi}} \tag{32}$$

$$At = \frac{\sum_{i=1}^{m} At_i \cdot \Delta t_{Ai}}{\sum_{i=1}^{m} \Delta t_{Ai}} \tag{33}$$

where Δt_{Oi} and Δt_{Ai} are the recording intervals of the operating and attitude parameters, respectively; Op_i and At_i are the operating and attitude parameters recorded at different time points during the short construction process, respectively; n and m are the amount of operating and attitude parameters recorded at this short construction process, respectively.

5.2. Data analysis and the load model validation

This section evaluates the predictive performance and reliability of the proposed load model using actual construction monitoring data, ensuring its applicability in real-world pipe-jacking scenarios.

5.2.1. Monitoring data analysis

The monitoring data collected during jacking Ring No. 11 of the left-line tunnel was analysed and used for verification of the established platform, based on which the horizontal and vertical moments ($\boldsymbol{M}(\boldsymbol{f}_{5z})$ and $\boldsymbol{M}(\boldsymbol{f}_{5y})$) induced by the thrust of articulation cylinder groups are calculated. The machine's position and the surrounding geological conditions are shown in Fig. 21, and the soil chamber pressure is detailed in the previous study [5]. The model validation primarily consists of two steps:

1. Based on the calculation process of the rectification moment shown in Fig. 20, the predicted moments (M_{yp}) for the validation section were computed by inputting soil parameters, operating parameters, and shield attitude into the prediction model. The soil parameters were determined based on the position of the pipe jacking machine, while the operating parameters (such as the soil chamber pressure) and shield attitude were derived from the preprocessed dataset.

2. The predicted moments (M_{yp}) for the validation section were compared with the measured moments (M_{ym}), and the prediction accuracy was evaluated using relevant error indexes.

To assess the accuracy of the platform's predictions, several error indexes were introduced, three of which consider absolute value, including Absolute Error (*AE*), Mean Absolute Error (*MAE*), Root Mean Squared Error (*RMSE*), and two consider relative value, including Relative Error (*RE*) and Mean Relative Error (*MRE*). These error indexes are defined as follows.

$$\begin{cases} AE = M_{yp} - M_{ym} \\ MAE = \frac{1}{N}\sum_{1}^{N} \left|M_{yp} - M_{ym}\right| \\ RMSE = \sqrt{\frac{1}{N}\sum_{1}^{N} \left(M_{yp} - M_{ym}\right)^2} \\ RE = \frac{\left|M_{yp} - M_{ym}\right|}{M_{ym}} \times 100\% \\ MRE = \frac{1}{N}\sum_{1}^{N} \frac{\left|M_{yp} - M_{ym}\right|}{M_{ym}} \times 100\% \end{cases} \tag{34}$$

where M_{yp} indicates the predicted moment forecasted by the platform; M_{ym} indicates the actual moment calculated by measured data; N represents the number of sample.

5.2.2. Validation of the established load model

Figures 22 and 23 illustrate the variations in horizontal and vertical moments as well as the associated error indexes. The figures show that the moments vary around their mean values. Both the predicted and measured moments exhibit a similar trend, with some discrepancies between predicted values and actual measurements, though these differences are not pronounced. The maximum relative errors for the horizontal and vertical moments for Ring No. 11 are 13.3 % and 4.4 %, respectively, with the vast majority of relative errors staying under the 10 % mark. In general, the predicted moments correspond closely with the measured data. However, discrepancies persist, primarily due to errors in soil parameter measurements, and monitoring data from the machine's guidance system. Table 6 presents the error indexes for the moments between predicted and measured data. The mean relative errors for horizontal and vertical moments in Ring No. 11 are 6.6 % and 1.7 %, respectively.

To assess the predictive stability of the model, the 95 % confidence intervals for the MAE and MRE have been obtained using the following formula:

$$\begin{cases} e_i = AE \text{ or } RE \\ \bar{e} = \frac{1}{N}\sum_{i=1}^{N} e_i \\ s_e = \sqrt{\frac{1}{N-1}\sum_{i=1}^{N} (e_i - \bar{e})^2} \\ CI = \bar{e} \pm z_{\alpha/2}\frac{s_e}{\sqrt{N}} \end{cases} \tag{35}$$

Where $z_{\alpha/2}$ indicates the corresponding quantiles of the t-distribution.

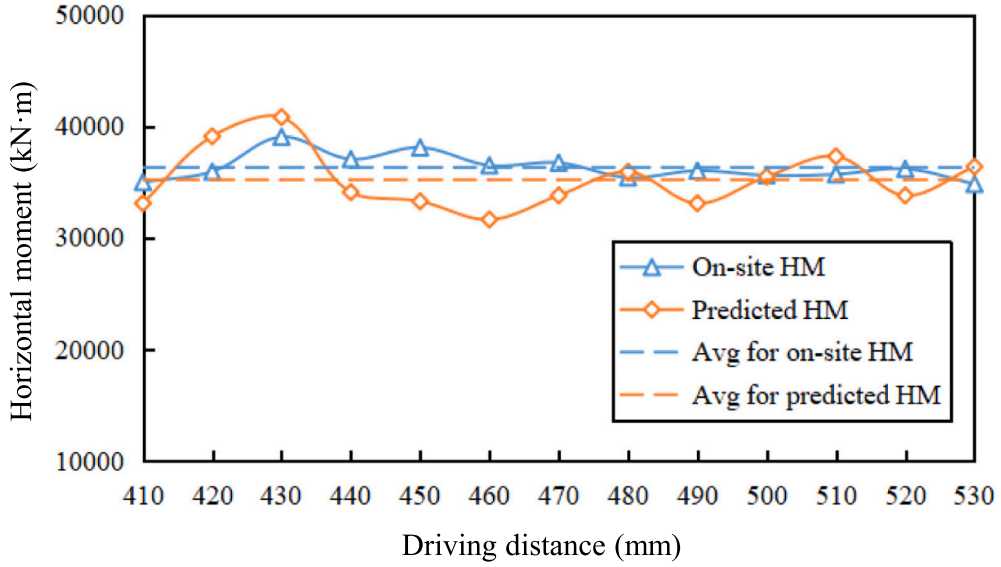
The confidence interval results are shown in Table 6. According to the statistical results, for horizontal moment prediction, the credible fluctuation range of the MAE is controlled around 3000 kN·m, and the credible fluctuation range of the MRE is controlled within 10 %. For vertical moment prediction, the credible fluctuation range of the MAE is controlled around 2000 kN·m, and the credible fluctuation range of the MRE is controlled within 3 %. The above confidence interval results effectively validate the predictive stability of the model. These results indicate that the platform's predictions are highly accurate and acceptable for practical engineering applications.

With the support of the platform, the horizontal deviation of the rectangular pipe-jacking machine during the jacking left-line tunnel was effectively controlled, as shown in Fig. 24. The horizontal deviation was maintained within the range of −20 to 40 mm, and the yaw angle was controlled within −4 to 2 mm/m.

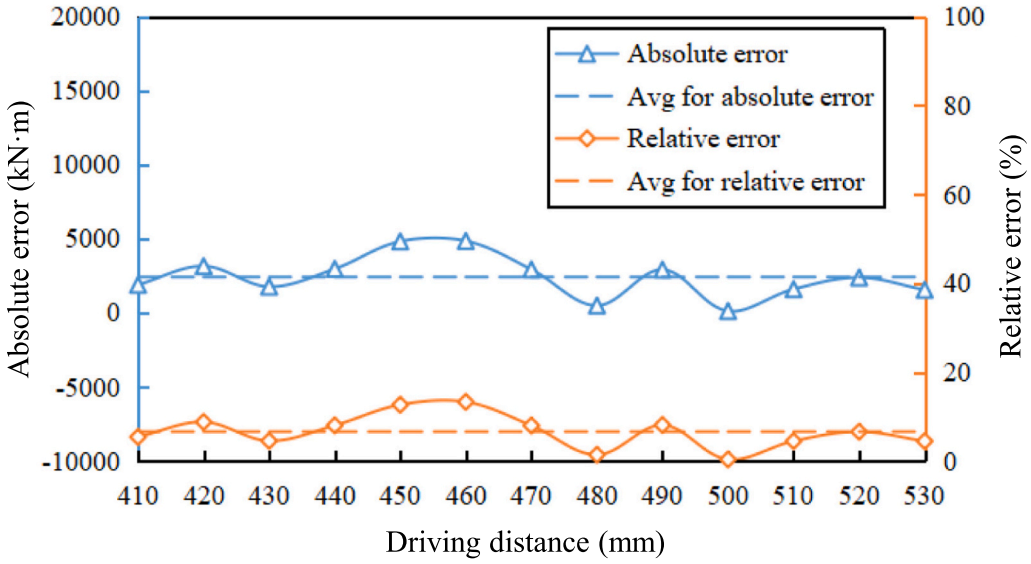
5.2.3. Risk warning analysis

The risk waring function of the established platform is shown in Fig. 25. At the start of jacking the left-line tunnel, the machine was right-handed, and the horizontal deviation approached 40 mm. Consequently, the machine was classified under a yellow risk warning level according to the platform's risk warning mechanism. Guidance for adjusting the machine's attitude was provided to the operator to make precise corrections and promptly reduce trajectory deviation. This adjustment ensured that the pipe-jacking machine steadily shifted to a left-handed position with a negative yaw angle. After continued adjustments of the machine's attitude, the risk level decreased to green as the jacking distance approached 30 m. Furthermore, the final horizontal deviation was −16 mm, significantly below the allowable deviation of ±50 mm.

Figure 26 shows the successful finishing of the case tunnel at Shasan



(a) Comparison of monitoring and predicted horizontal moments



(b) Error evaluation

Fig. 22. Horizontal moment and error evaluation.

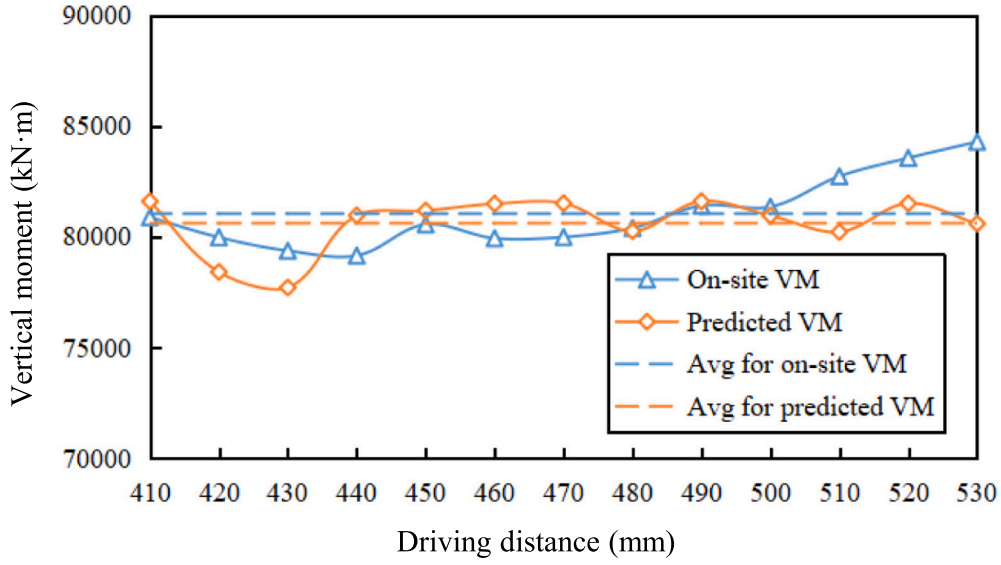
Station, with the pipe-jacking machine's trajectory deviation meeting construction safety control requirements. During this project, the established intelligent attitude control platform effectively supported real-time risk warning and intelligent decision-making for attitude correction, thereby facilitating high-quality rectangular pipe jacking.

5.3. Challenges and solutions

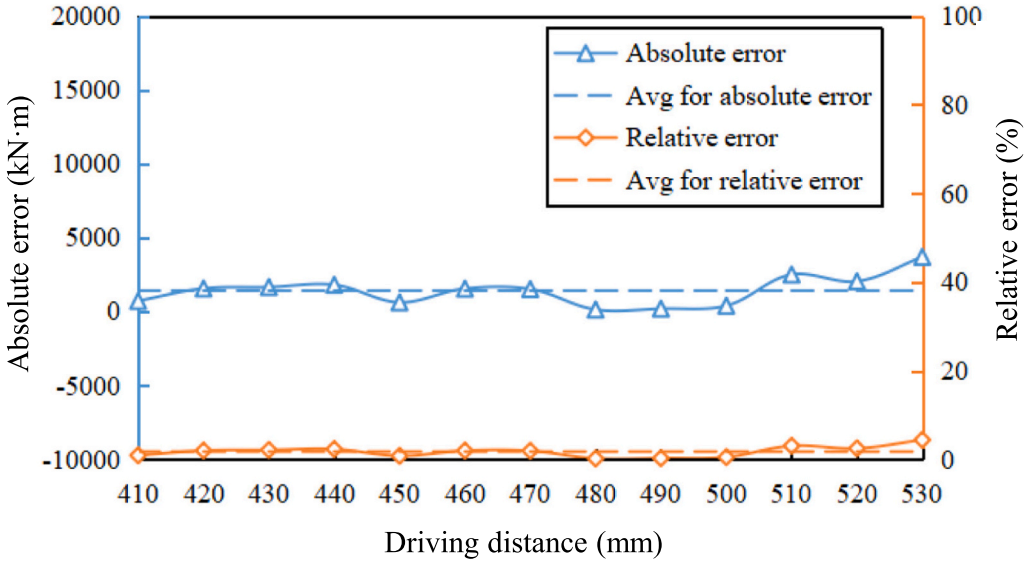
Despite the platform's demonstrated effectiveness, its practical deployment faces several technical challenges. This section outlines the major difficulties encountered and the corresponding strategies adopted to address them.

5.3.1. Integration of multiple subsystems

The digital twin platform requires seamless coordination among multiple hardware and software systems, particularly in integrating sensors, control systems, and data transmission platforms. Ensuring the interoperability of these components while maintaining platform stability and reliability presents a significant challenge. To tackle this issue, a modular system architecture has been adopted to optimize system integration. A phased testing approach ensures that each subsystem operates stably before full-scale integration. Additionally, high-performance middleware has been introduced to facilitate efficient and seamless data communication across different systems, ensuring smooth and reliable information exchange.



(a) Comparison of monitoring and predicted vertical moments



(b) Error evaluation

Fig. 23. Vertical moment and error evaluation.

Table 6
Error indexes of Ring No. 11 of the case tunnel.

Error index	Horizontal moment	Vertical moment
MAE (KN·m)	2426	1420
RMSE (KN·m)	2782	1723
MRE	0.066	0.017
CI-MAE	[1575 3276]	[8122028]
CI-MRE	[0.043 0.089]	[0.010 0.024]

5.3.2. Data processing and visualization

The attitude control of the pipe-jacking machine involves processing large volumes of data. Ensuring the accuracy of data acquisition, processing, and feedback while minimizing platform latency remains a technical challenge. In response to this challenge, a cloud-edge-terminal computing architecture has been employed, distributing data processing tasks across on-site devices. This reduces the burden on remote servers and minimizes latency. Furthermore, optimization of data transmission pathways, coupled with data compression techniques, ensures the stability and timeliness of real-time data streams. To enhance user experience, a layered display approach and high-efficiency graphical rendering techniques have been implemented, prioritizing key information and enabling users to quickly and accurately access critical machine status data.

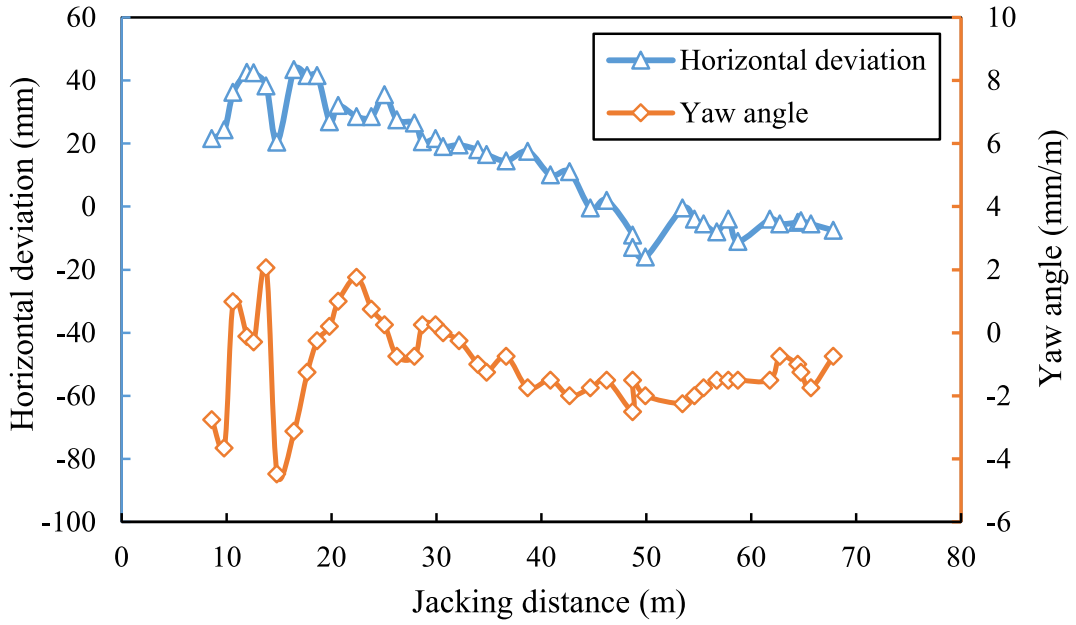


Fig. 24. Horizontal deviation and yaw angle of the machine.

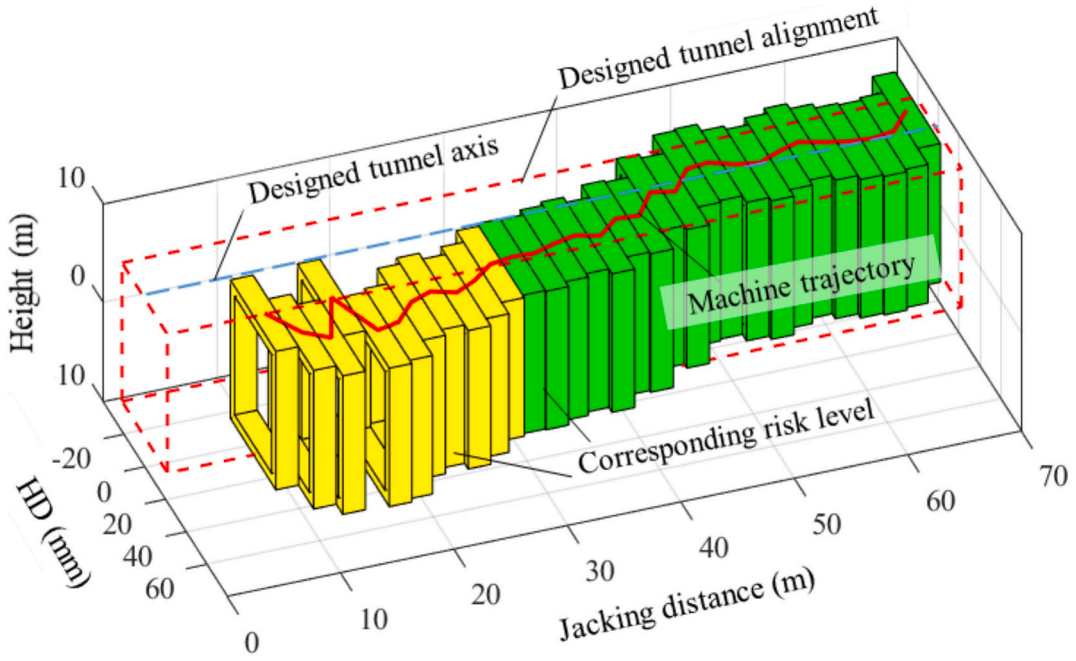


Fig. 25. Trajectory of pipe-jacking machine and risks evaluation.

5.3.3. Reliability assessment of sensor data

During pipe-jacking operations, sensor data may be affected by improper installation or disturbances from cutters rotation, leading to measurement errors that are difficult to identify programmatically. As these errors accumulate, they can negatively impact attitude control and correction accuracy. To ensure the reliability of real-time correction decisions, regular inspections and calibration of sensor data must be conducted throughout the construction process. This approach helps mitigate the impact of measurement errors and enhances the overall stability and accuracy of attitude control.

6. Conclusions and future research

This paper proposed a mechanics-informed digital twin (DT) framework for real-time intelligent attitude control of rectangular pipe-jacking machines, aiming to reduce trajectory deviation and improve construction safety and precision. The framework was successfully applied and validated in the Shasan Station project in Shenzhen, China.

The results demonstrate that the proposed DT-based intelligent control platform significantly improves trajectory correction accuracy. By integrating a theoretical load model that captures soil–machine interaction characteristics, the platform achieved high-precision prediction of correction moments, with most prediction errors below 10 %. As a result, the final horizontal deviation of the pipe-jacking machine



(a) Jacking construction of the case tunnel (b) Reception of the pipe-jacking machine

Fig. 26. Completion of the case tunnel.

was controlled within −16 mm, far better than the allowable ±50 mm. These findings validate the model's predictive accuracy and highlight its practical effectiveness in complex underground environments.

Compared with traditional experience-based or purely data-driven methods, this approach combines mechanical principles with real-time data processing, reducing operator dependency and enhancing response efficiency. It offers a globally applicable solution for intelligent control in large-section pipe-jacking projects, particularly under heterogeneous and sensitive geological conditions.

Future work should focus on enhancing the adaptability of the DT model through hybrid physics–machine learning algorithms and extending its functions to include real-time face stability assessment and ground deformation prediction [42]. Further integration of intelligent alert systems with multimodal outputs could support more comprehensive risk management during tunnelling operations.

CRediT authorship contribution statement

Dong Su: Project administration, Conceptualization. **Jianhang Chen:** Writing – original draft. **Penglu Chen:** Investigation. **Lei Wang:** Writing – review & editing, Writing – original draft. **Jinwei Zhou:** Visualization. **Xiangsheng Chen:** Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Data availability

Data will be made available on request.

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